



PSYCH 201B

Statistical Intuitions for Social Scientists

Modeling Dependent Data: Linear Mixed Models

Welcome to our LAST week together 🥹🎉

1. Today: Review & LMM Essentials
2. Tues: Challenge Review, Final Lab, & Proposal Feedback (we'll have grades up by the evening)
3. Wed: Course-Wrap Up & Final Project Time

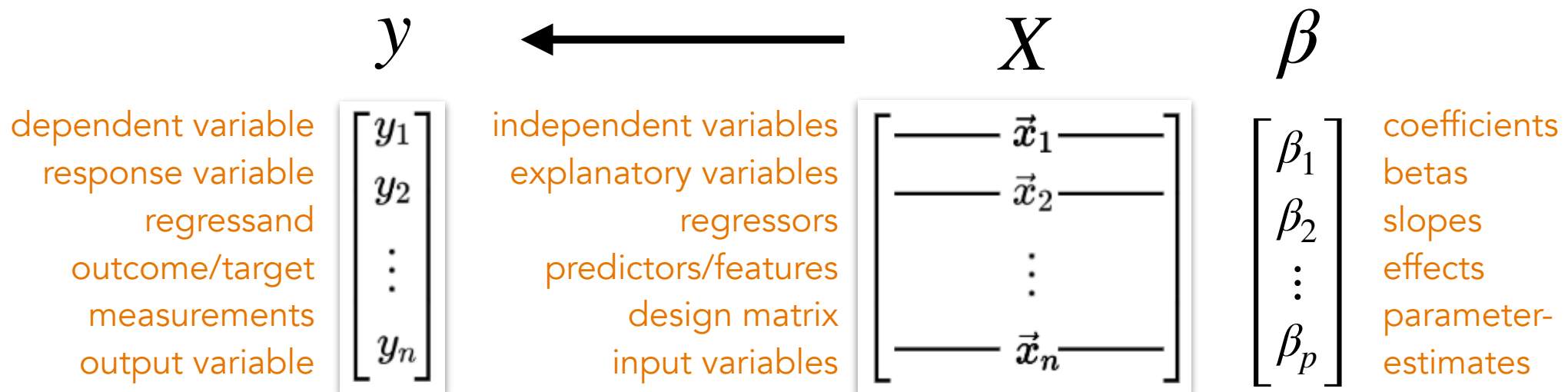
Note

The last date to `commit` and `push` any updates to previous labs, HWs, and challenges for mastery-based grading is **March 20th by midnight**.

Review: General Linear Model (GLM)

$$y = X\beta + \epsilon$$

Models a **scalar** variable **y** as a linear function of predictor **variables X**



Regression(s) are just “flavors” of GLM

$$y = X\beta + \epsilon$$

Simple (univariate) regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \epsilon_i$$

one predictor

one column wide

**technically 2 with intercept*

y

dependent variable
response variable
regressand
outcome/target
measurements
output variable

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$



independent variables
explanatory variables
regressors
predictors/features
design matrix
input variables

X

$$\begin{bmatrix} \text{---} \vec{x}_1 \text{---} \\ \text{---} \vec{x}_2 \text{---} \\ \vdots \\ \text{---} \vec{x}_n \text{---} \end{bmatrix}$$

β

coefficients
betas
slopes
effects
parameter-
estimates

$$\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

Regression(s) are just “flavors” of GLM

$$y = X\beta + \epsilon$$

Multiple regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi} + \epsilon_i$$

many predictors

many columns wide

y

dependent variable
response variable
regressand
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measurements
output variable

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$



independent variables
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Regression(s) are just “flavors” of GLM

$$y = X\beta + \epsilon$$

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi} + \epsilon_i$$

categorical (switches)

t-test, ANOVA, ANCOVA, etc

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dependent variable
response variable
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Regression(s) are just “flavors” of GLM

$$y = X\beta + \epsilon$$

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi} + \epsilon_i$$

binary
logistic regression
(classification)



y

dependent variable
response variable
regressand
outcome/target
measurements
output variable

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$p_i = \frac{1}{1 + e^{-(X\hat{\beta})}}$$



independent variables
explanatory variables
regressors
predictors/features
design matrix
input variables

$$\begin{bmatrix} \text{---} \vec{x}_1 \text{---} \\ \text{---} \vec{x}_2 \text{---} \\ \vdots \\ \text{---} \vec{x}_n \text{---} \end{bmatrix}$$

Logistic link function
transform linear predictor to
expected probability [0, 1]

X

β

$$\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

coefficients
betas
slopes
effects
parameter-
estimates

Regression(s) are just “flavors” of GLM

$$y = X\beta + \epsilon$$

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi} + \epsilon_i$$

counts

poisson regression
(chi-square tests)



y

dependent variable
response variable
regressand
outcome/target
measurements
output variable

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\mu_i = e^{X\hat{\beta}}$$



independent variables
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regressors
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design matrix
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Log link function
transform linear predictor to
expected count

X

$$\begin{bmatrix} \text{---} \vec{x}_1 \text{---} \\ \text{---} \vec{x}_2 \text{---} \\ \vdots \\ \text{---} \vec{x}_n \text{---} \end{bmatrix}$$

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Review: GLM Assumptions

Errors should be independent & identically **distributed (iid)**

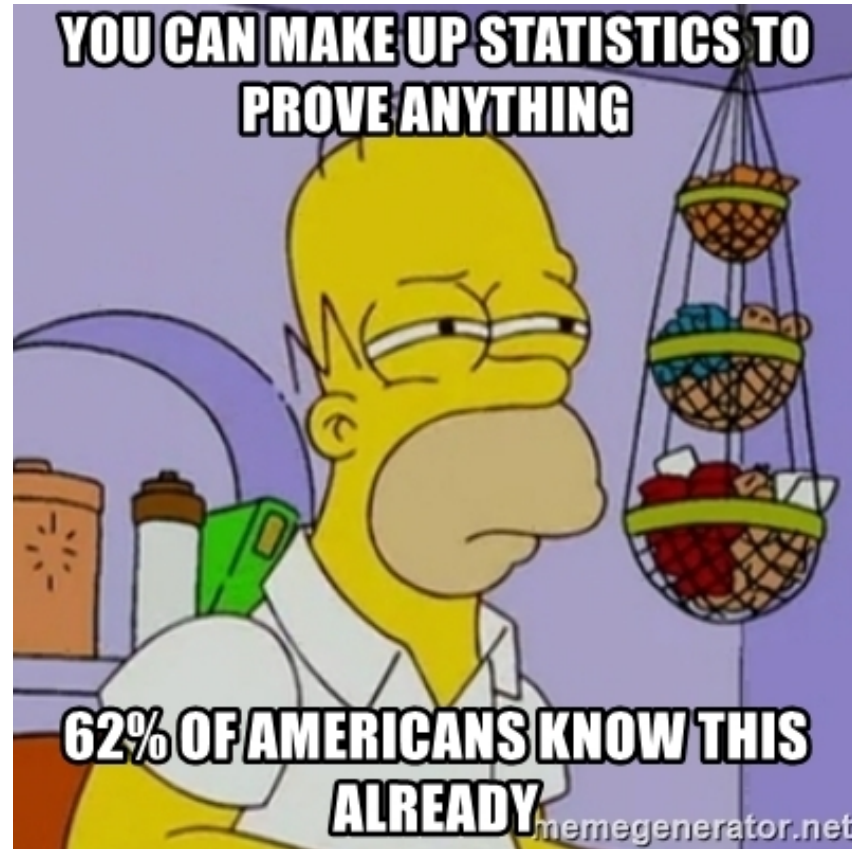
- **Normality:** assumes that the *residuals* are normally distributed. It's actually okay if the predictors and the outcome are non-normal, so long as the residuals are normal.
- **Additivity & Linearity:** assumes that the outcome **y** is a linear function of separate predictors **X** that are added together
- **Homoscedasticity:** assumes that the variance of our residuals doesn't change as function of our predictors; model errors shouldn't depend upon value of predictor
- **No perfect multi-collinearity:** assumes that our predictors **X** are not just linear combinations of each other; can't figure out what the *unique variance* each one explains
- **Independence of errors:** assumes our residuals don't depend upon each other - this gets violated when you have **repeated-measures, time-series, geospatial, or multi-level data**

Violating Independence

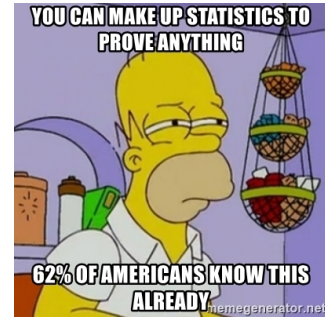
- So far, all the GLM flavors you've seen satisfy the **iid assumption**
- But you'll often work with data where it's violated, e.g:
 - **psychology experiments**: repeated measurements from the same individual
 - **survey data**: multiple measurements from the same city/county/state
 - **time series**: observation at time-point t depends upon $t - 1$

What's the big deal?
What could go wrong?

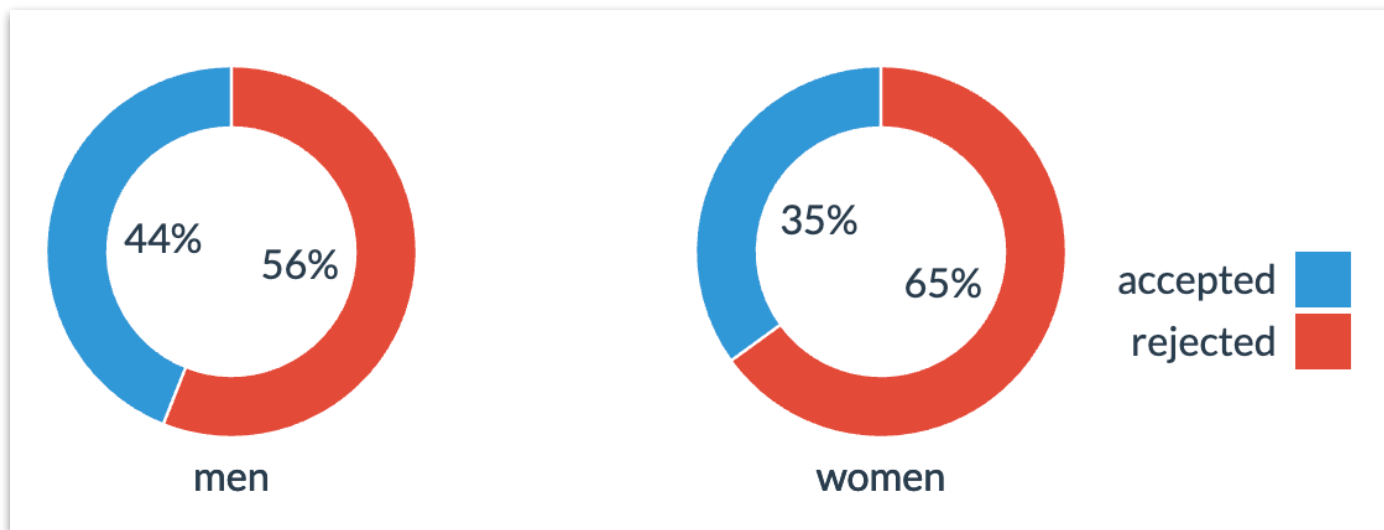
Simpson's paradox



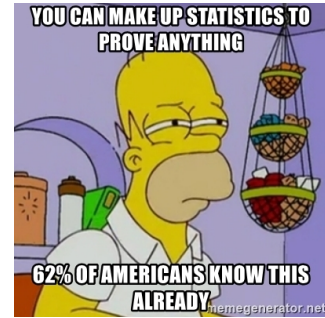
Simpson's paradox



In 1973, the University of California-Berkeley was almost sued for sex discrimination. The numbers looked pretty incriminating: the graduate schools had just accepted 44% of male applicants but only 35% of female applicants.



Simpson's paradox

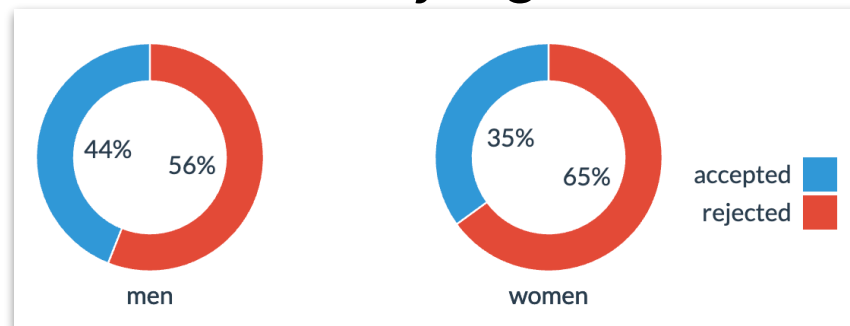


In 1973, the University of California-Berkeley was almost sued for sex discrimination. The numbers looked pretty incriminating: the graduate schools had just accepted 44% of male applicants but only 35% of female applicants.

When researchers looked at the evidence, though, they uncovered something surprising:

What?

*If the data are **properly pooled** by department... there is a small but statistically significant **bias in favor of women***



Simpson's paradox

Aggregated by Sex only

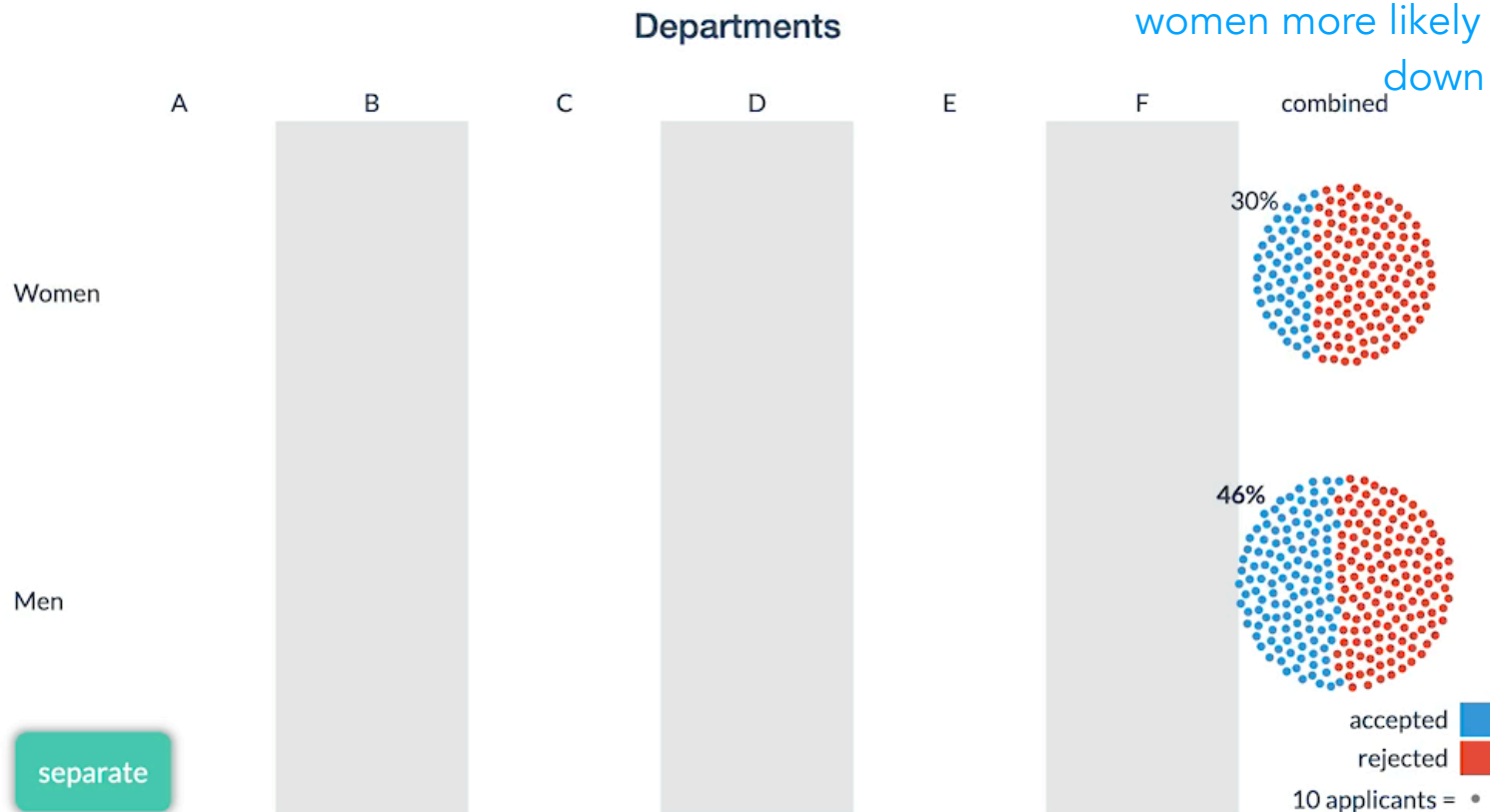
	Men		Women	
	Applicants	Admitted	Applicants	Admitted
Total	8442	44%	4321	35%

overall men are more likely to be admitted

Aggregated by Sex & Dept

Department	Men		Women	
	Applicants	Admitted	Applicants	Admitted
A	825	62%	108	82%
B	560	63%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%

women more likely to be admitted when broken down by department



Simpson's paradox

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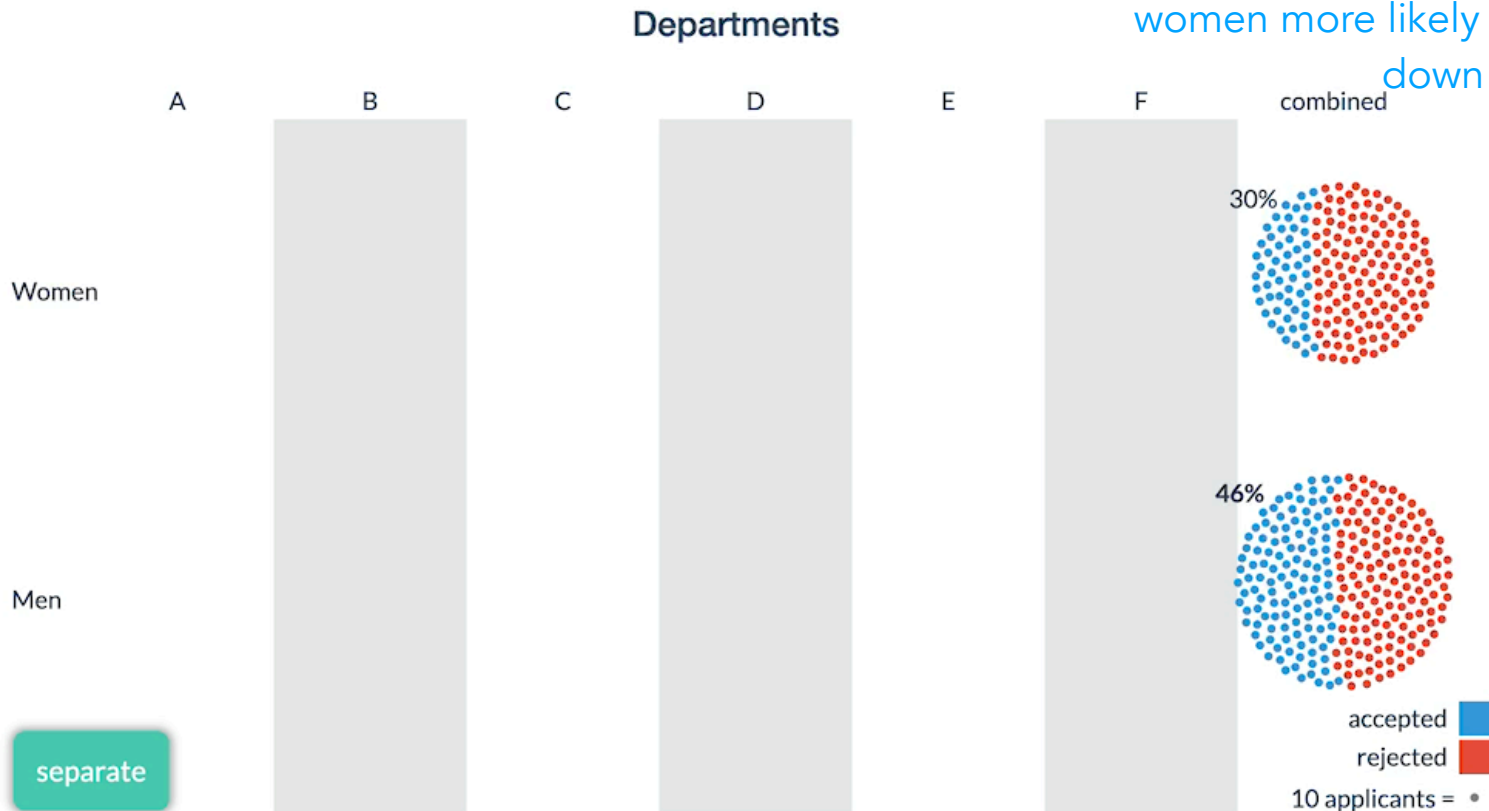


Ignores variance between departments!

Aggregated by Sex & Dept

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women more likely to be admitted when broken down by department



Women applied to more competitive departments!

Simpson's paradox

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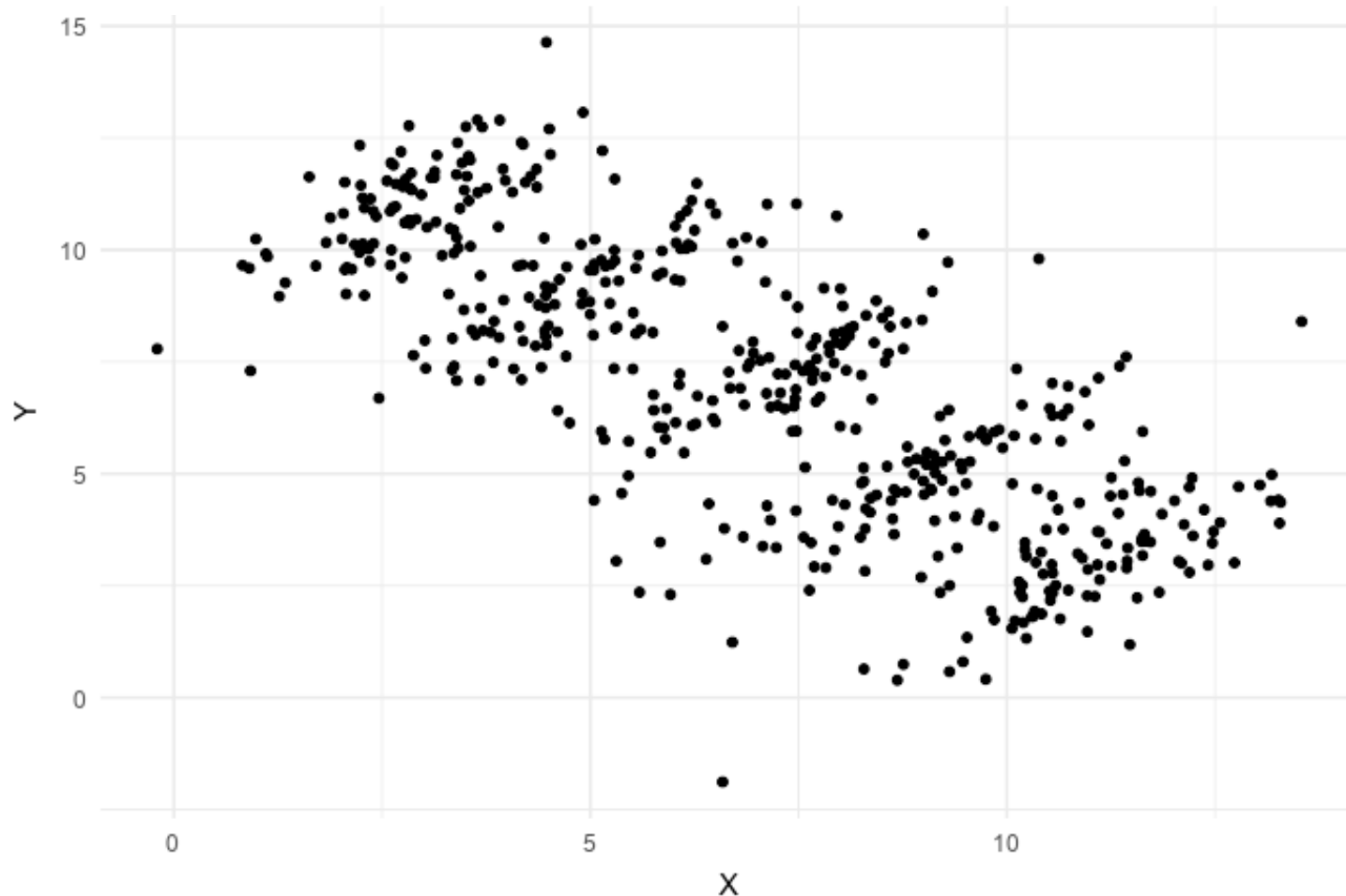
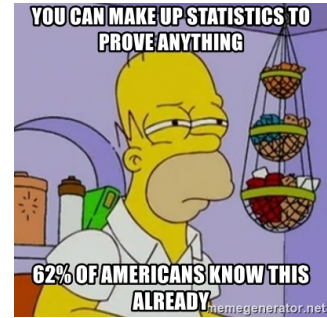
In fact, the pooled and corrected data showed a "small but statistically significant bias in favor of women."



Bickel, P. J., Hammel, E. A., & O'Connell, J. W. (1975). Sex Bias in Graduate Admissions: Data from Berkeley. *Science*, 187(4175), 398-404.

Simpson's paradox

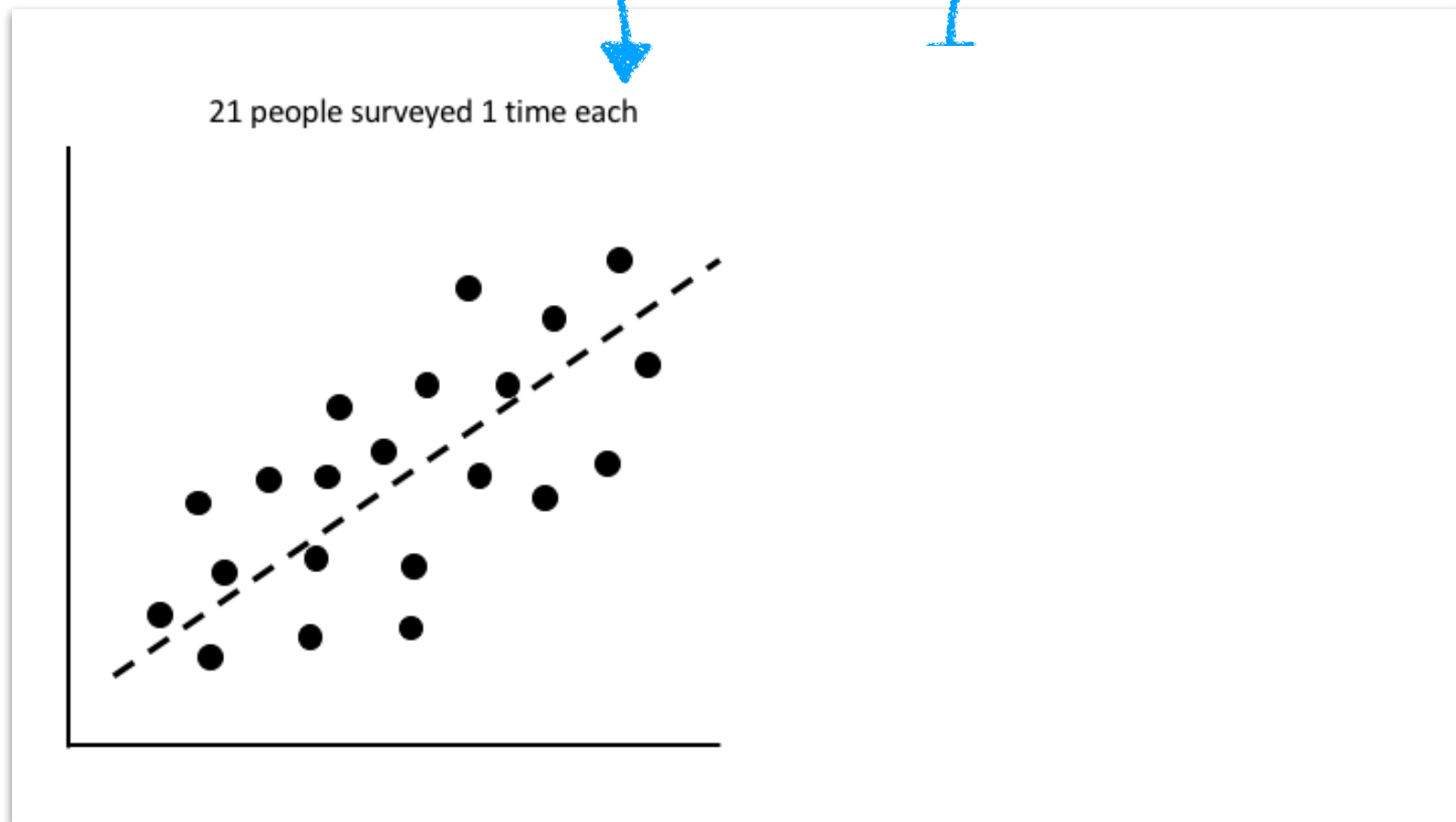
Phenomenon in statistics when a trend that appears in several **groups** of data disappears or reverses when the groups are **combined**



Dependent Data

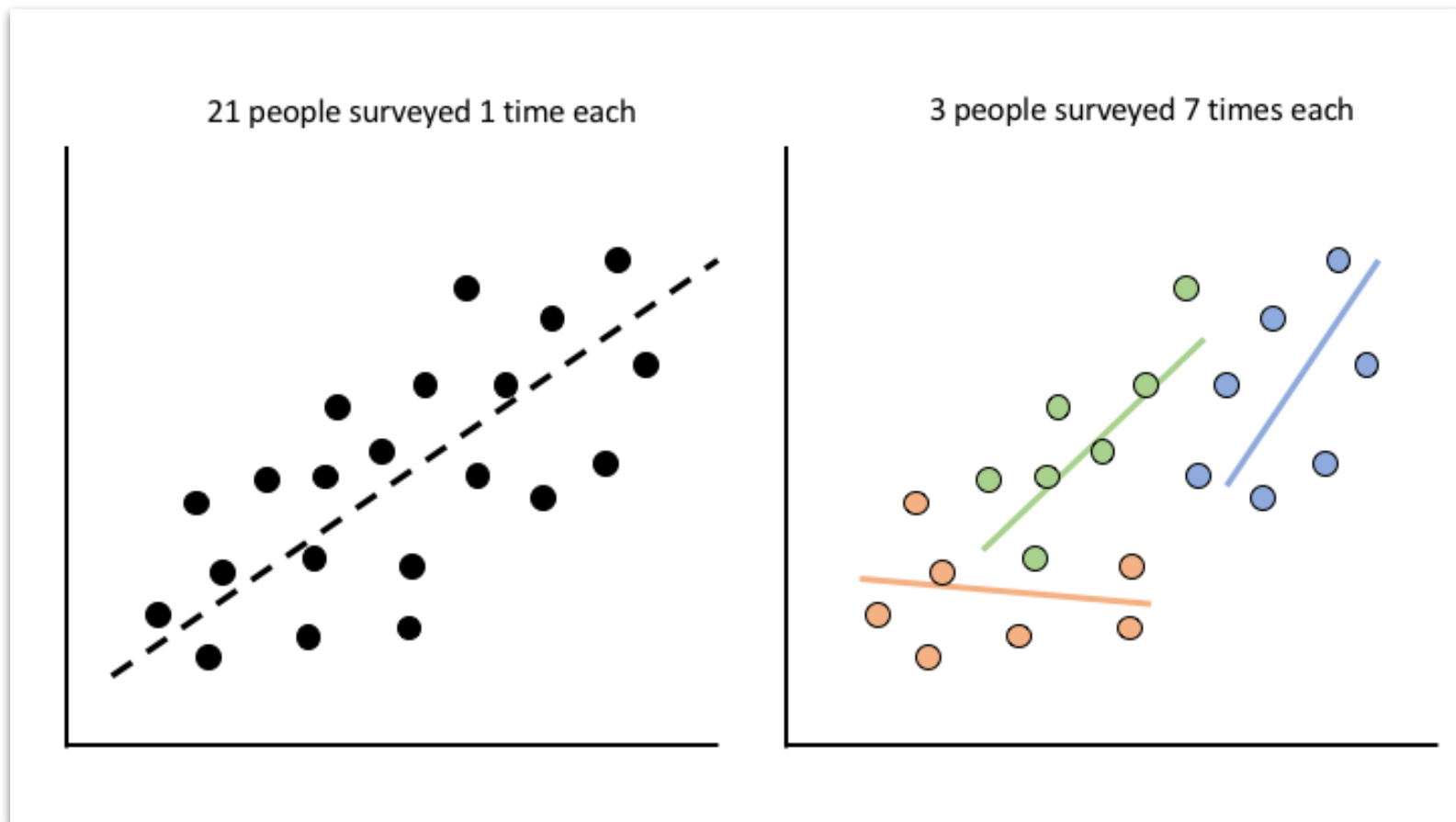
Each person's response is independent of everyone else's response

Data points *within* each person are more similar than data points across people



Perils of Ignoring Dependence

- Parameter estimates could be **inaccurate** or **even reversed**
- **Standard-errors too small** = t-stats too big = p-values too small = our inferences are too liberal!
- **Higher Type I error** (remember which one that is?)

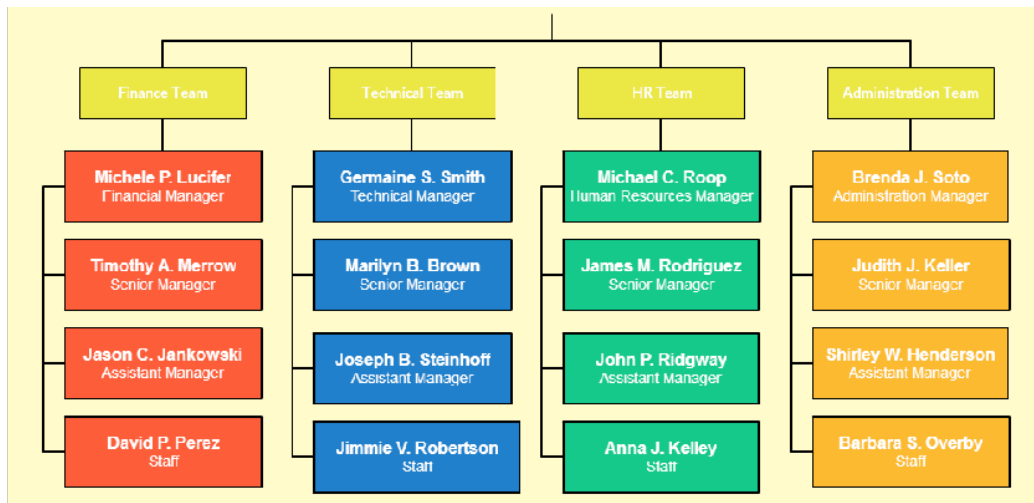


The solution: Linear Mixed Models

An extension of GLM that allows us to
account for dependencies in data by
estimating parameters at different *levels*

Hierarchical data

e.g. person *within* team *within* company

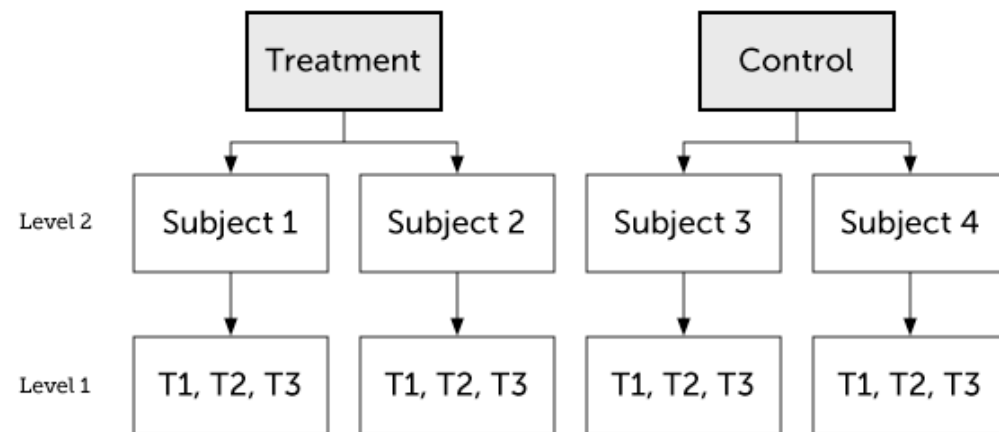


department > applicant

state > county > city

Longitudinal data

e.g. repeated measures per person



experimental condition > subject > trial

group > subject > time-point

The solution: **Linear Mixed Models**

An extension of GLM that allows us to
account for dependencies in data by
estimating parameters at different levels

Wait what about repeated-measures ANOVA?

Stop using them

- Anything you can do with rm-ANOVA you can do with LMMs!
- With rm-ANOVA we know our model is misspecified (doesn't handle iid violation) so we try to correct it after-the-fact: "sphericity violation" Greenhouse-Geisser correction
- LMMs let us explicitly model dependence in different ways
 - And if you really need F-ratios, you *already know* how to get them (nested model comparison applies to LMMs too!)

The solution: **Linear Mixed Models**

It is a shame that psychology students don't really learn much about analyzing multilevel data. Think about studies you have read recently in psychology or neuroscience. How many of them take a single measurement on the DV from each participant? Very few, if any. Nearly all take multiple measurements, for one or more of the following reasons:

- (1) the researchers are measuring the same participants across levels of a factor in a within-subject design;*
- (2) they are interested in assessing change over time;*
- (3) they are measuring responses to multiple stimuli.*

*Multilevel data is so common that **multilevel analysis should be taught as the default approach in psychology.***

*Learning about multilevel analysis can be challenging, but you already know much of what you need by having learned about regression...**you will see that it is just an extension of simple regression.***



Dale J Barr
(big deal in LMMs & language)

Linear mixed effects models

Linear mixed effects models*

**Often taught as a full quarter class by itself!*

We'll try to cover the essential intuitions today 😊

Terminology & Jargon

- Linear Mixed Models are also called...
 - **H**ierarchical **L**inear **M**odel (HLMs)
 - **M**ulti-**L**evel **M**odel (MLMs)
 - Mixed Effects model
 - Random Effects model
 - Variance Components model
 - Growth Curve model

Different names for the same approach!

LMMs: The Essentials

1. Ways to aggregate dependent data
 - Complete, No-pooling, **Partial-pooling**
2. Parts of the mixed model
 - Fixed vs Varying Effects
 - What's happening under-the-hood
3. Specifying Varying Effects
 - Random Intercepts & Slopes
 - How to choose your varying effects

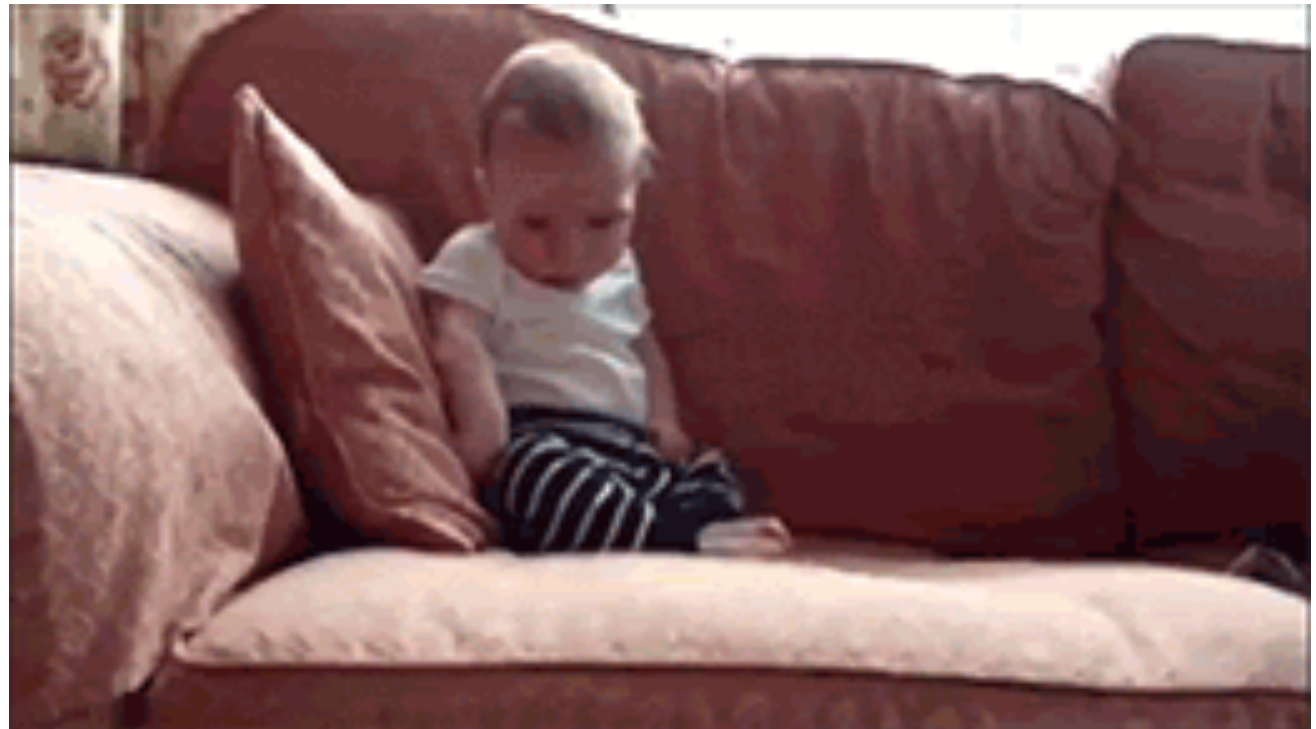
Let's build intuitions with a real dataset...

1. Ways to aggregate dependent data
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Dataset

How does sleep deprivation affect reaction time?

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
309	0	222.73
309	1	205.27
309	2	202.98
309	3	204.71
309	4	207.72

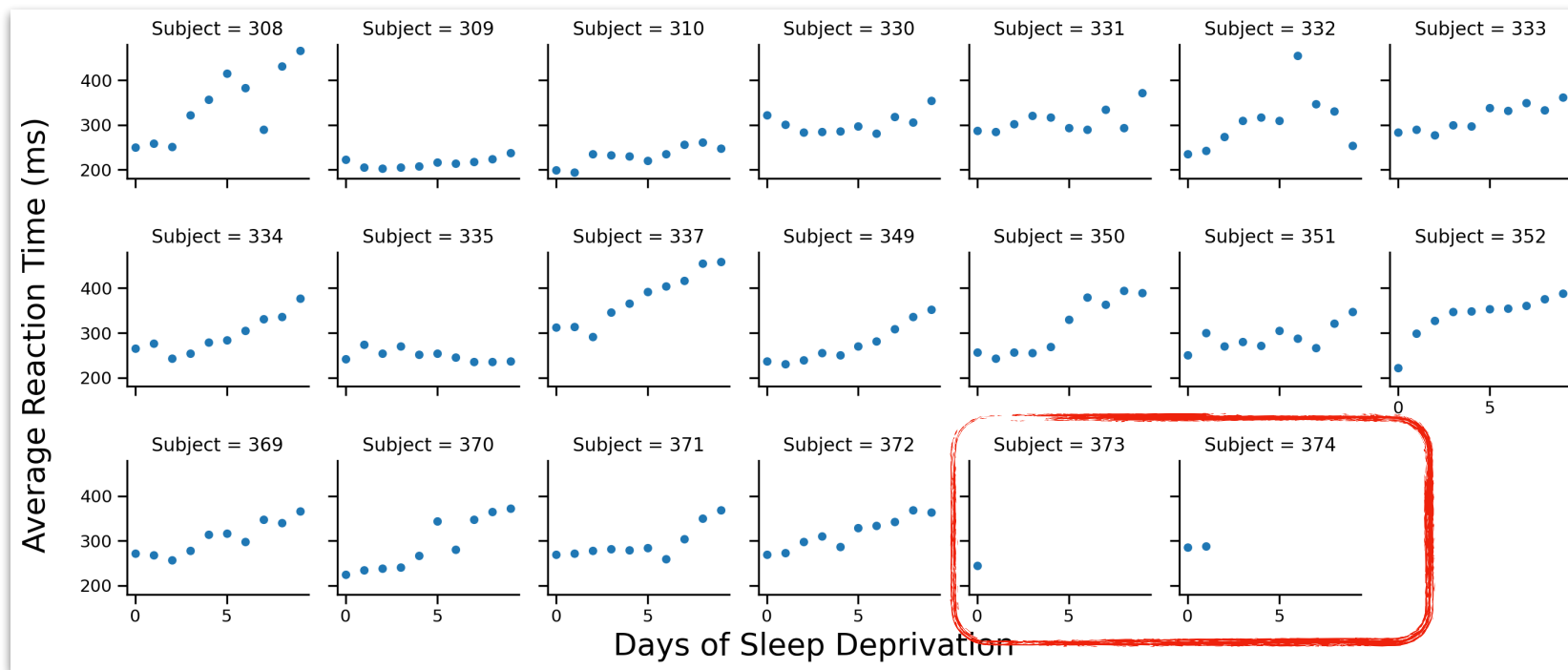


Dataset

How does sleep deprivation affect reaction time?

20 participants; 5 measurements per participant

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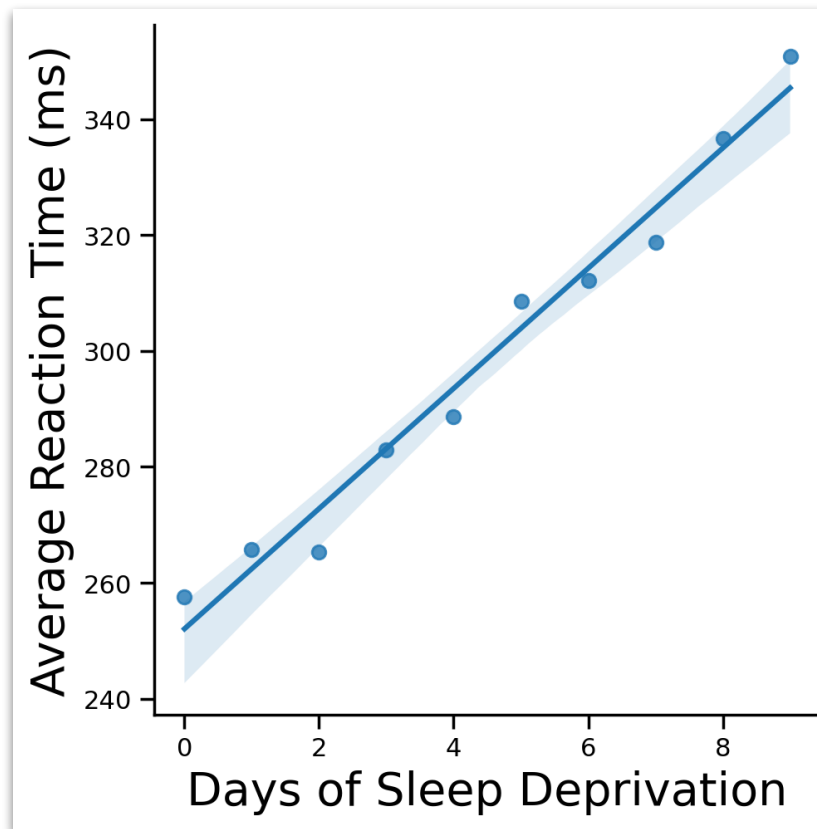
2 with incomplete information

Ways to NOT to Aggregate dependent data

Averaging over replicates: average participants within day and run a regression on the averages

Avoid this whenever possible

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308	0	249.56
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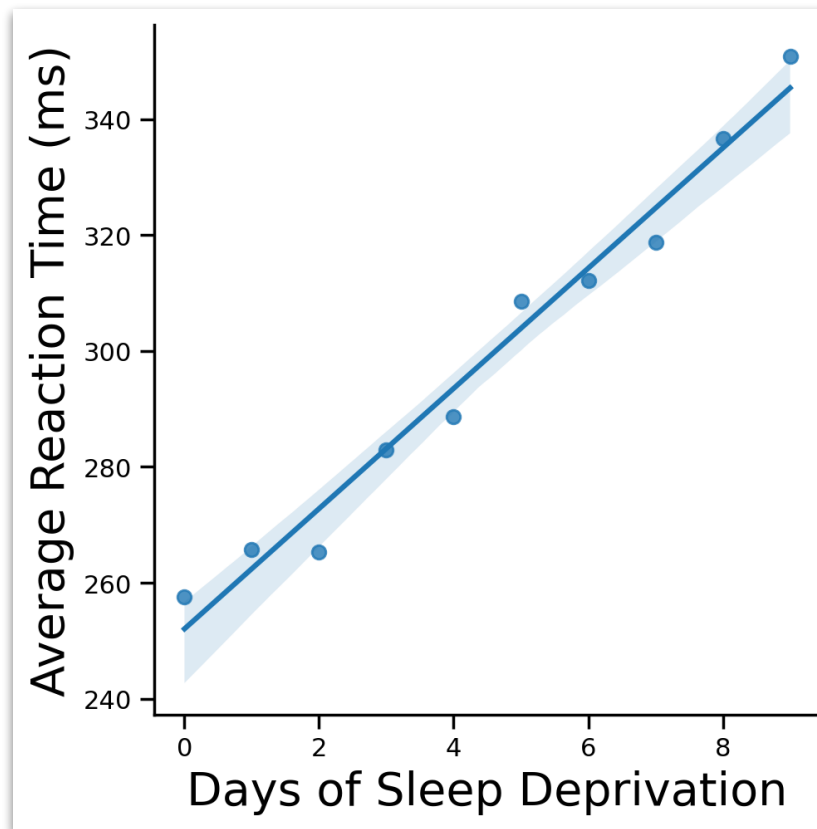
Unnecessarily throws away useful information about variance within people!

Ways to NOT to Aggregate dependent data

Averaging over replicates: average participants within day and run a regression on the averages

Avoid this whenever possible

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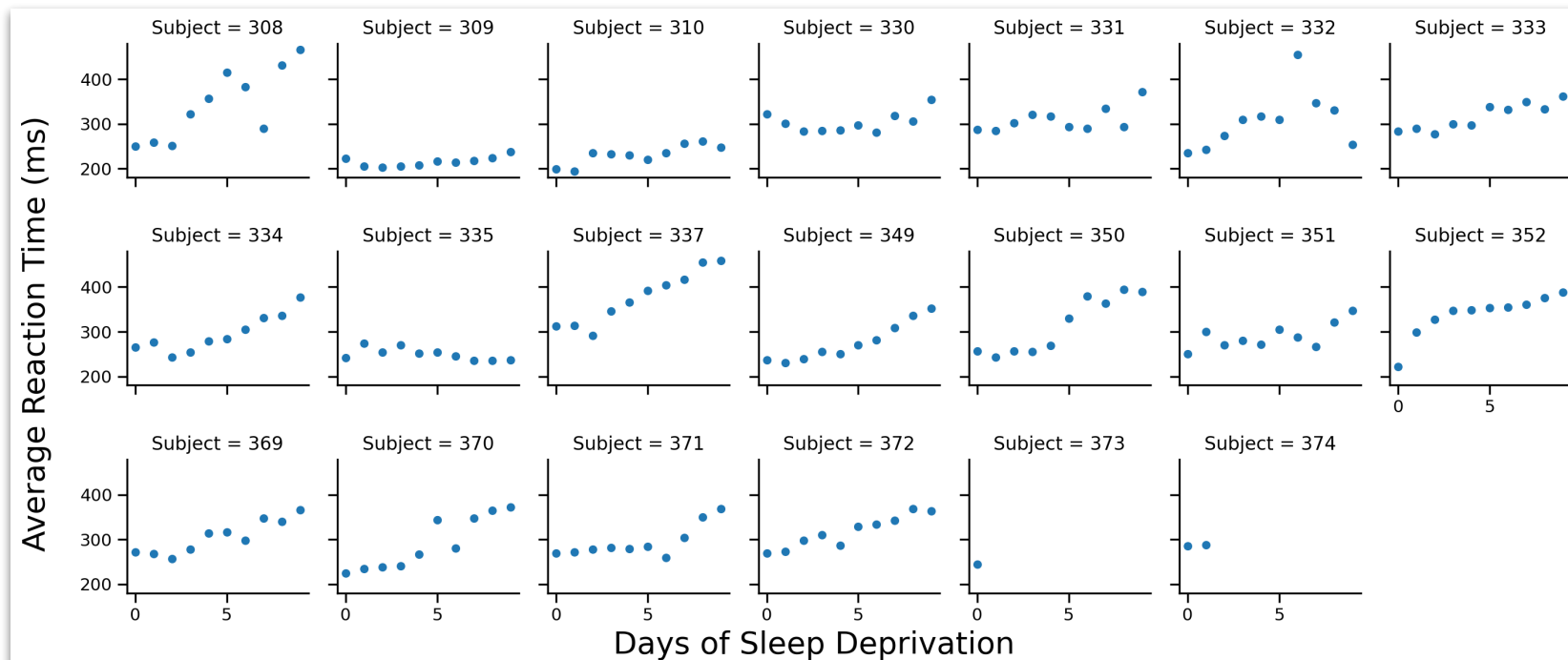
Ok we did do this...for cats dataset by averaging over *trials* within each cat...but just to simplify things!

1. Ways to Aggregate dependent data

Complete pooling: combine data from all participants (ignore dependence) and fit one **global GLM**

20 participants; 5 measurements per participant

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308	0	249.56
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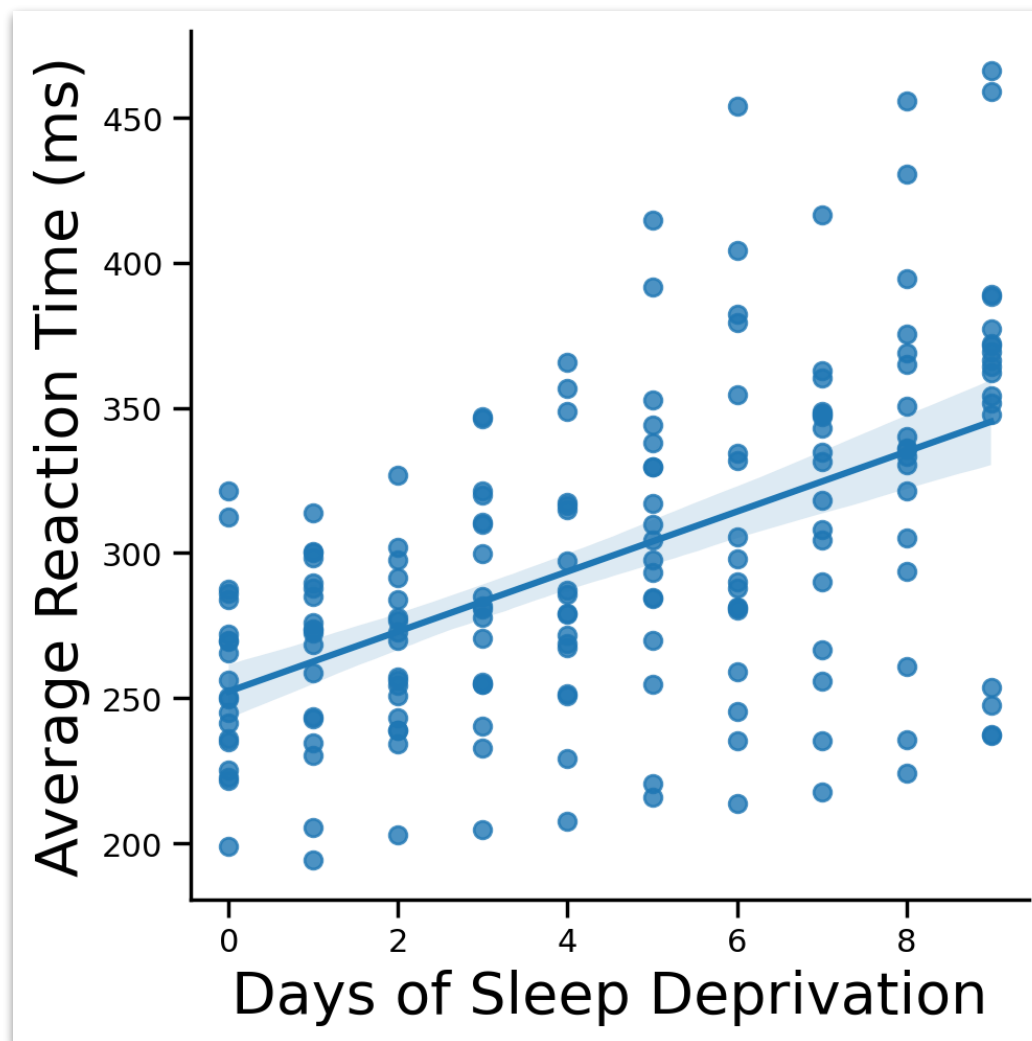


2 with incomplete information

1. Ways to Aggregate dependent data

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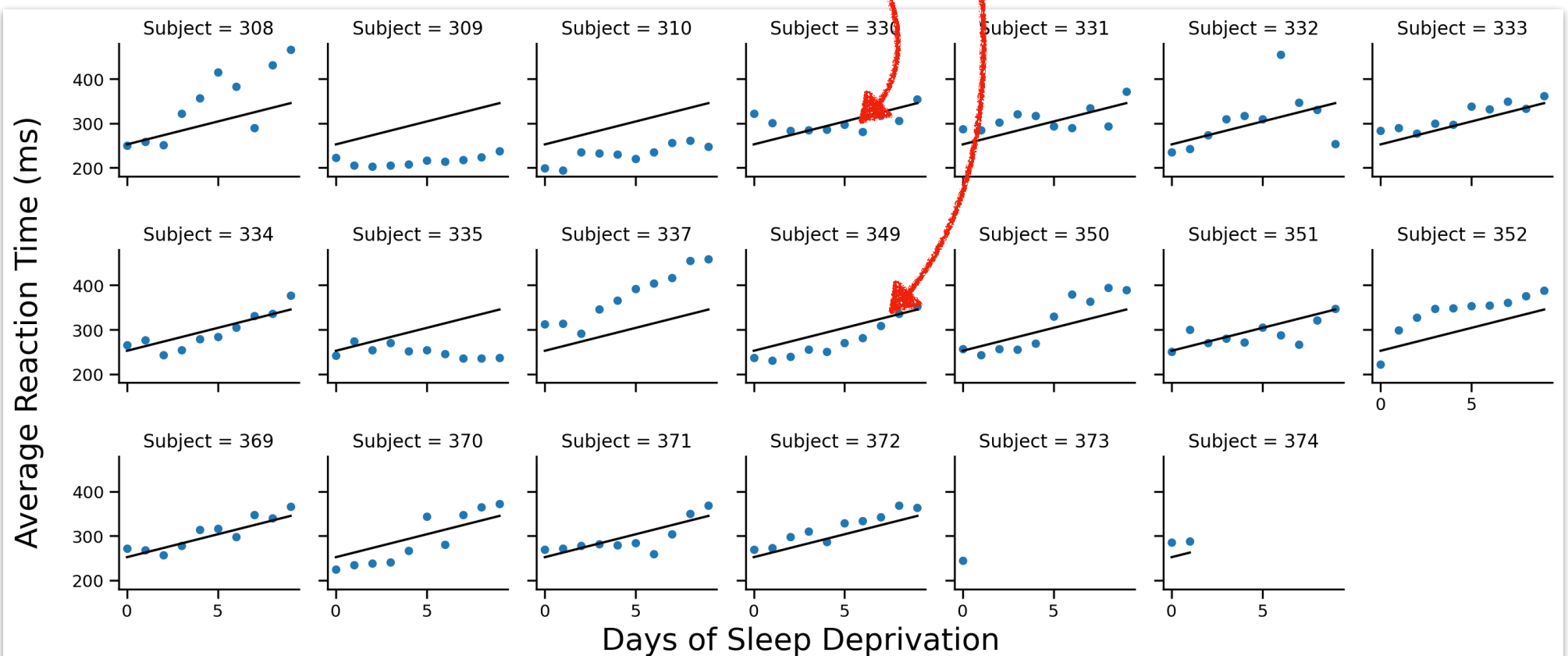
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Complete Pooling: Fit 1 global GLM

```
model("Reaction ~ Days", data = sleep).fit()
```

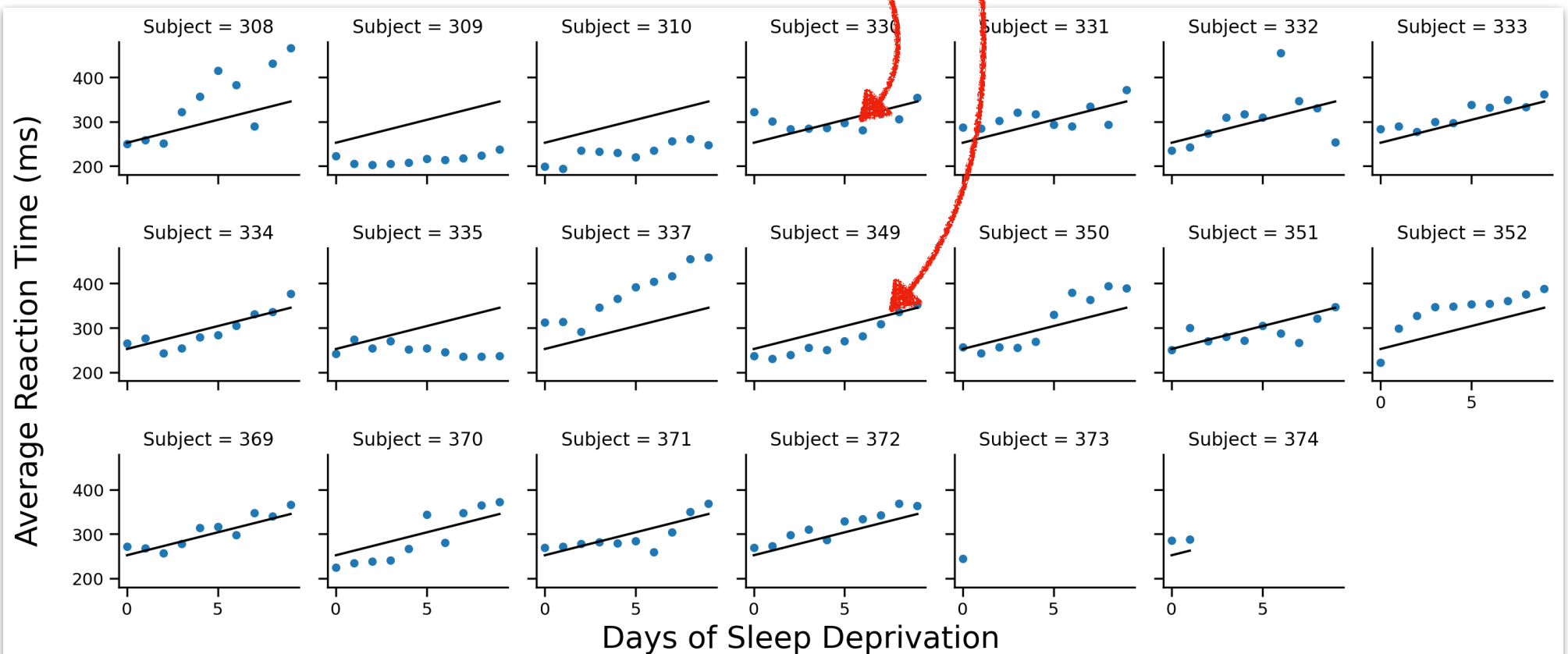
Same regression intercept and slope
for every Subject



Complete Pooling: Fit 1 global GLM

Ignores variability across Subjects!

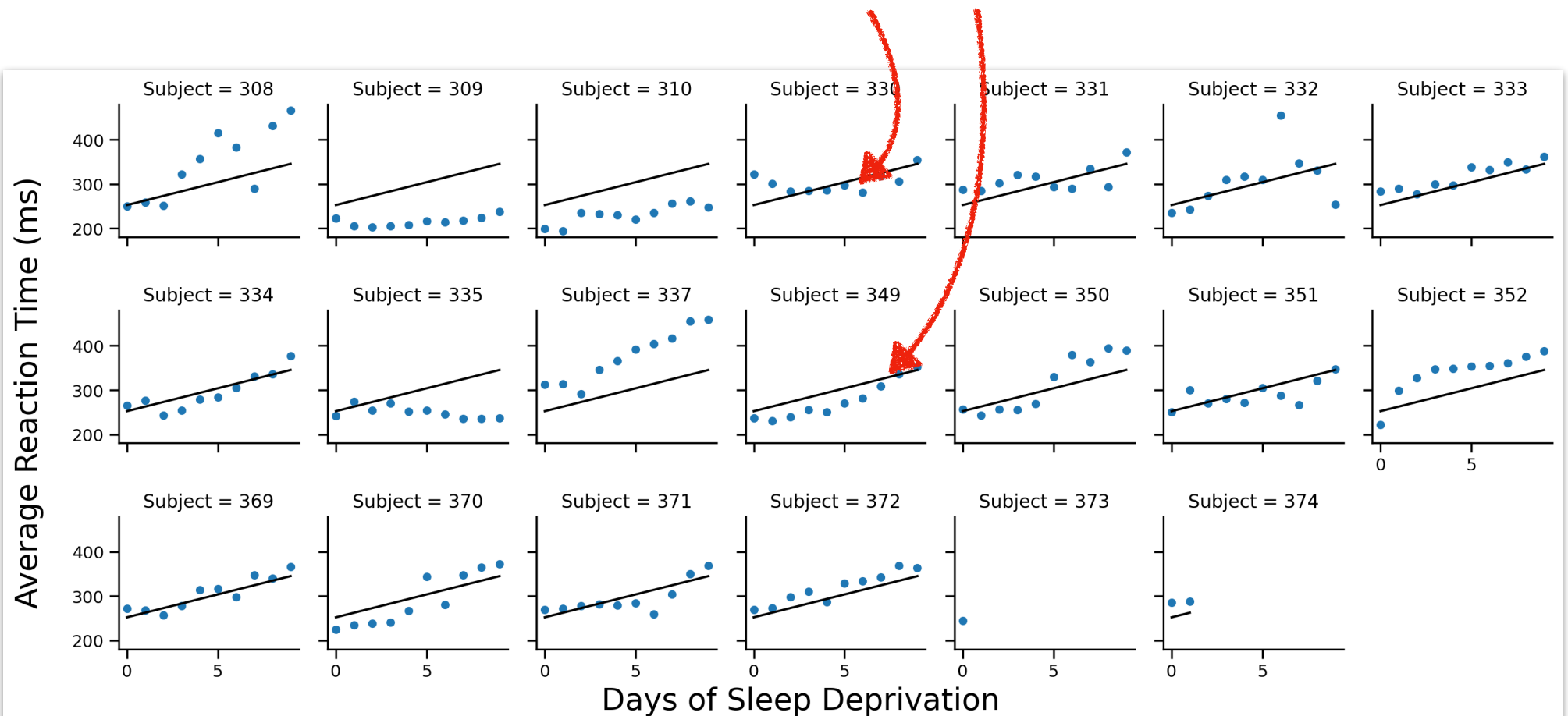
Same regression intercept and slope
for every Subject



Complete Pooling: Fit 1 global GLM

Ignores variability across Subjects!
Risks Simpson's Paradox!

Same regression intercept and slope
for every Subject

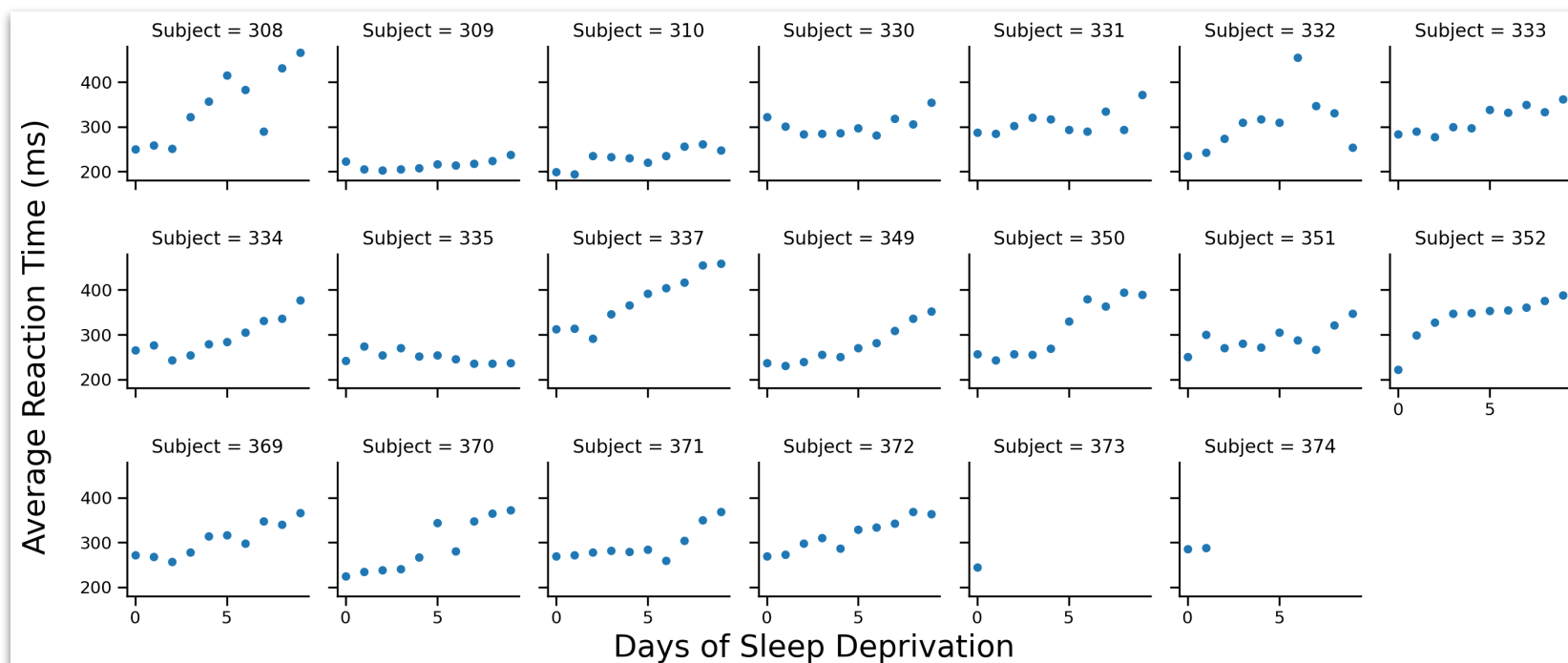


2. Ways to Aggregate dependent data

No pooling: don't combine any data and fit 20 separate **GLMs**, one for each Subject

20 participants; 5 measurements per participant

subject	days	reaction
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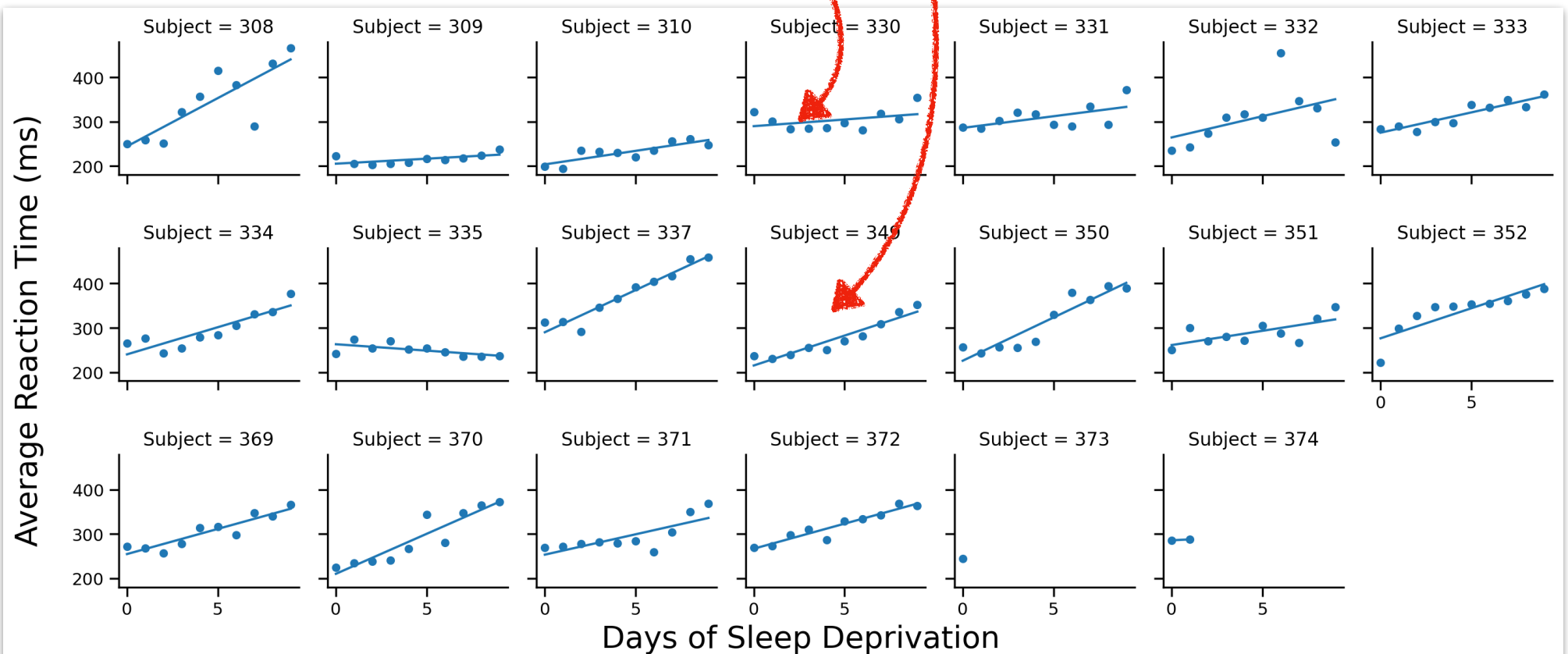


2 with incomplete information

No Pooling: Fit separate GLMs

```
for sub in sleep["Subject"].unique():  
    sub_data = sleep.filter(col("Subject") == sub)  
    model("Reaction ~ Days", data = sub_data).fit()
```

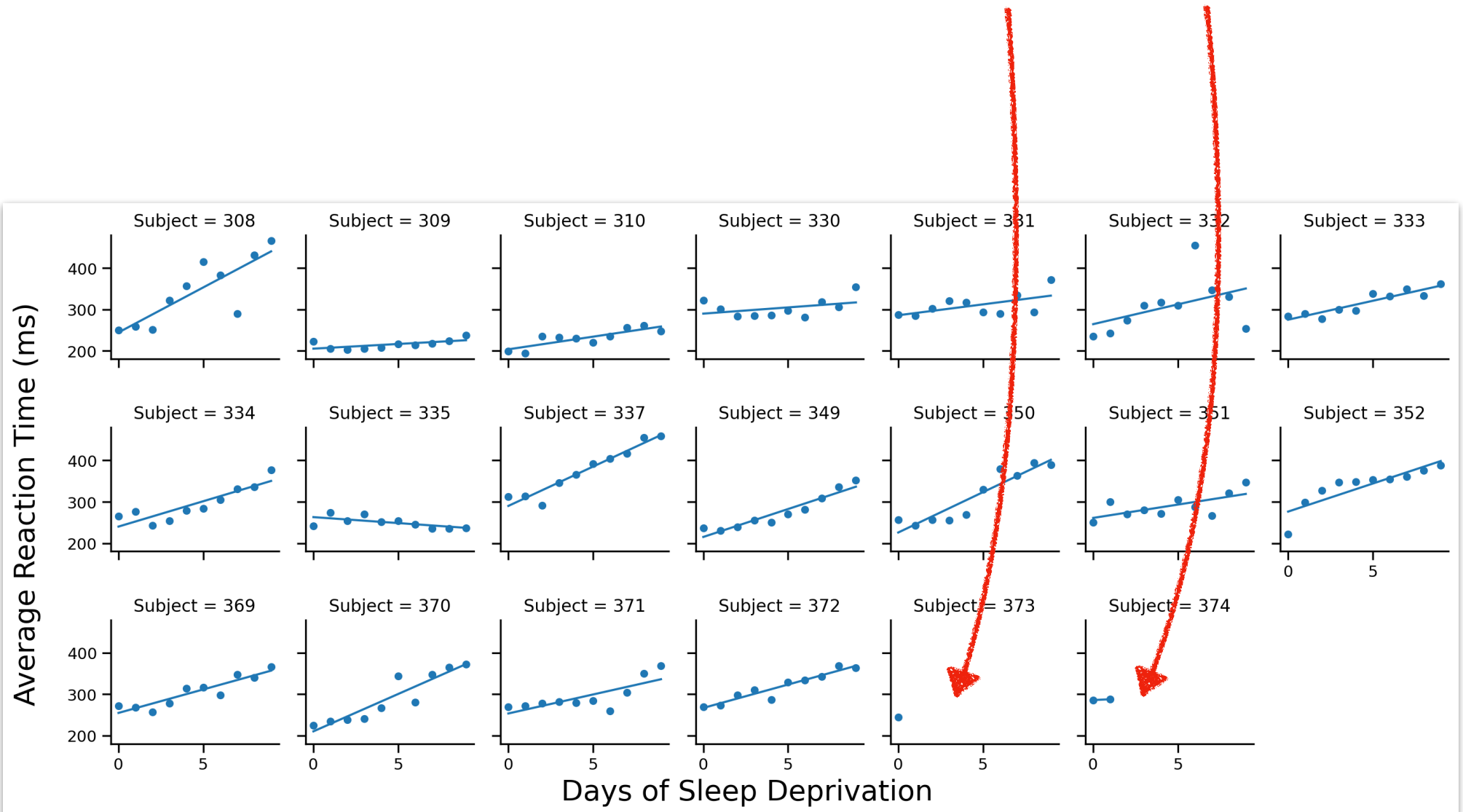
Independent regression intercept and slope
for every Subject



No Pooling: Fit separate GLMs

Treats each subject as a **separate dataset**

Poor/inestimable models when too little/missing data

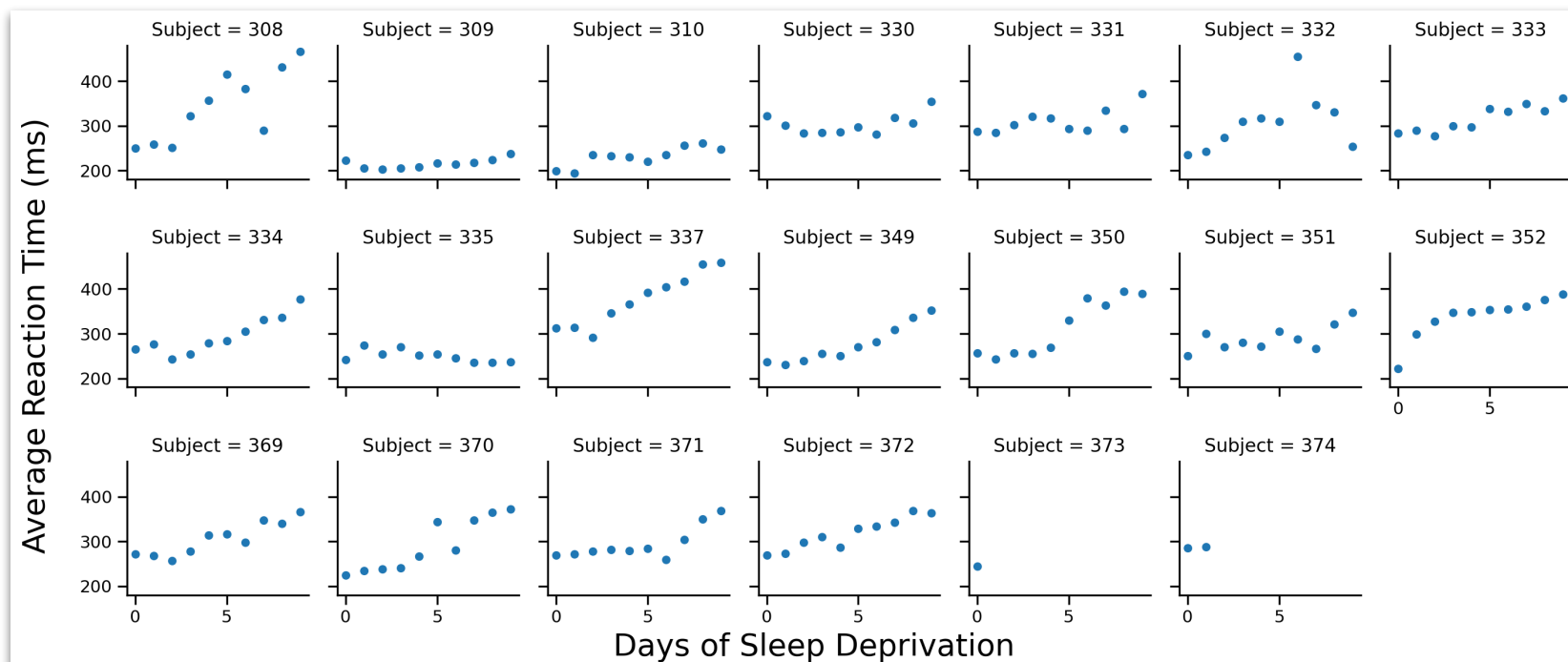


3. Ways to Aggregate dependent data

Partial pooling: take into account all data by **explicitly modeling variation** between participants

20 participants; 5 measurements per participant

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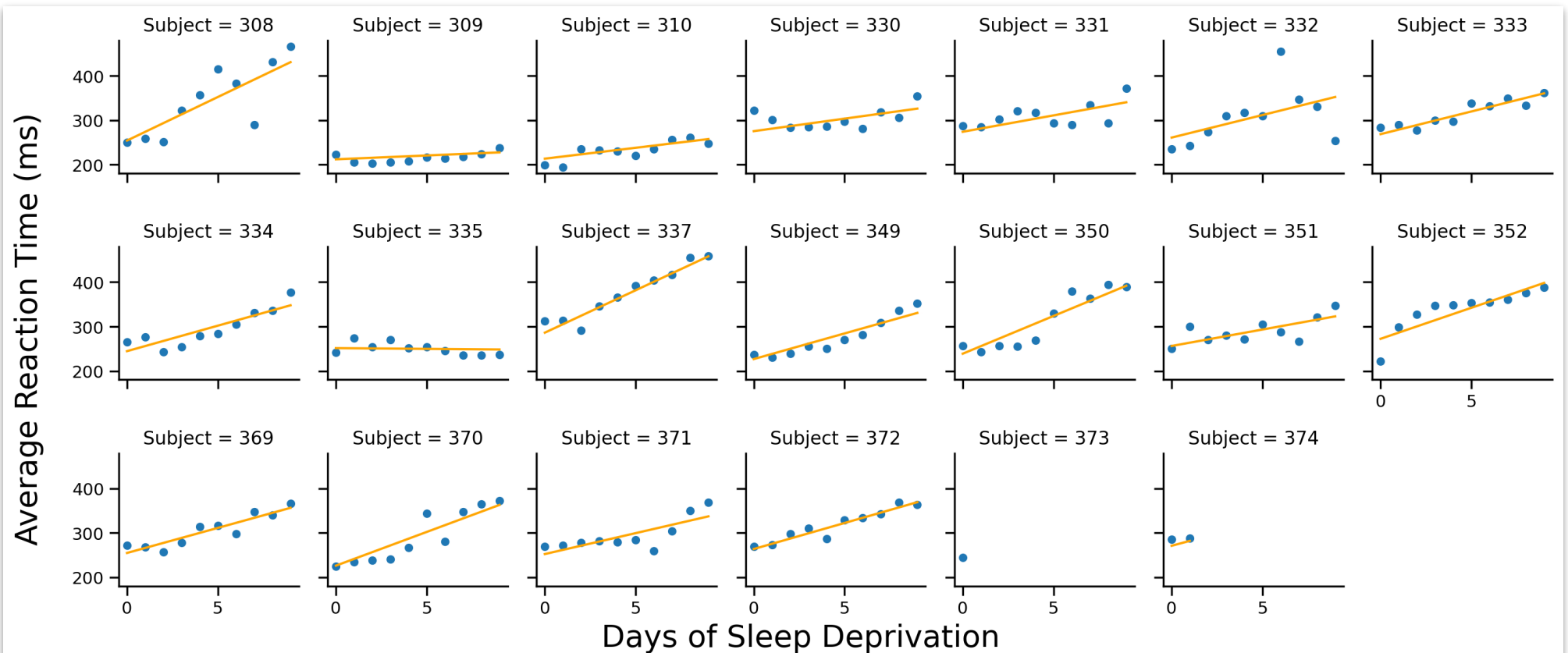
Partial Pooling: Fit (G)-Linear Mixed Model

`model("Reaction ~ Days + (1 + Days | Subject)",
data = sleep).fit()`

intercept slope

how to model dependence

Each Subject is assumed to **independently deviate**
from a global **Intercept & Slope**



Ways to Aggregate dependent data

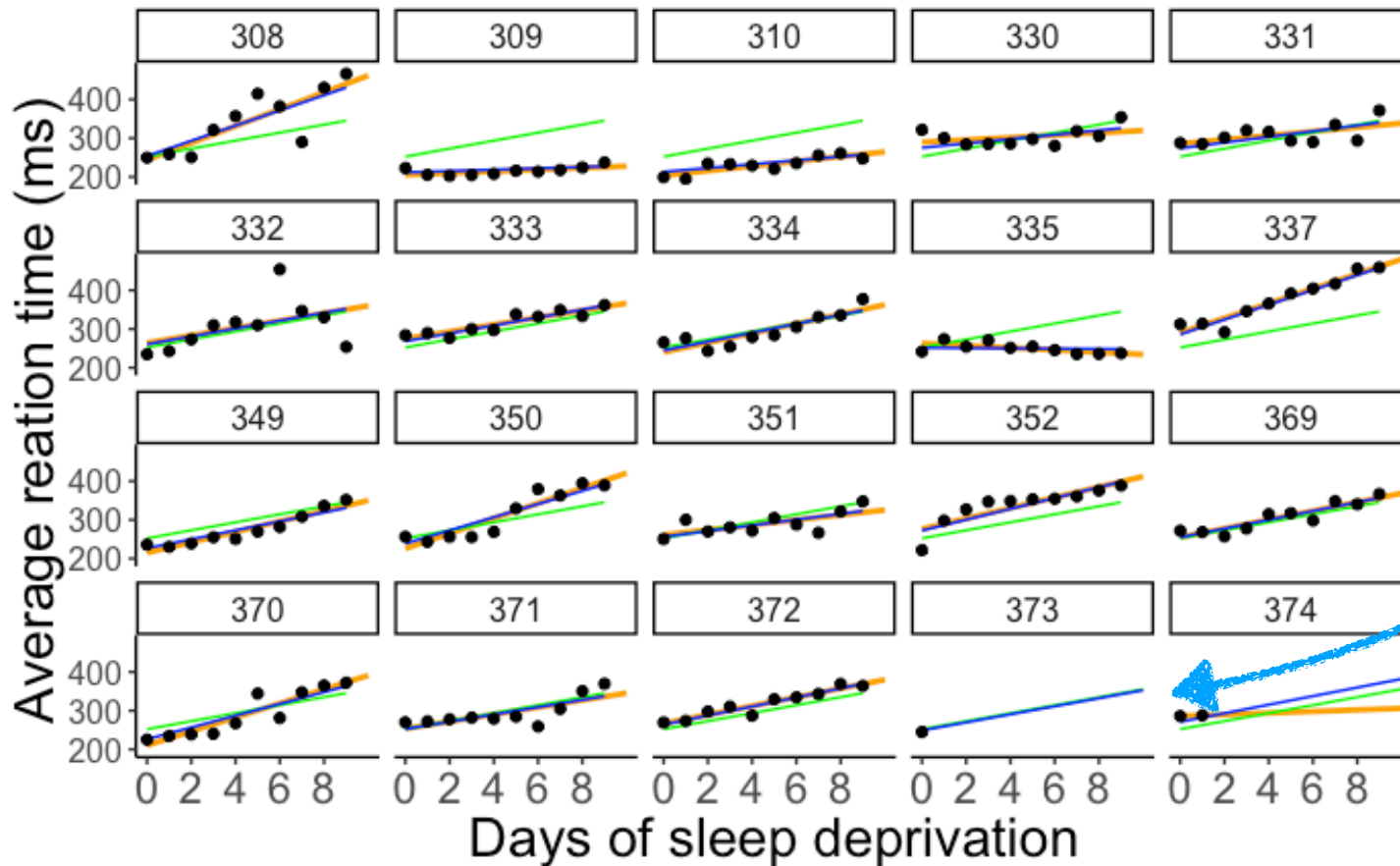
1. complete pooling Doesn't account for individual variation

2. no pooling

No/noisy predictions with one/few observations
Doesn't account for shared/general effect across Ss

3. partial pooling

Explicit model of multiple sources of variation
Extrapolation possible when data is missing



LMMs: The Essentials

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 - What's happening under-the-hood
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Linear *Mixed* Model: What's getting Mixed?

**Every field agrees these are terrible names*

Fixed Effects

Varying (Random) Effects

- **Why I don't use the term "fixed and random effects"**

Posted on [January 25, 2005 6:41 PM](#) by [Andrew](#)

People are always asking me if I want to use a fixed or random effects model for this or that. I always reply that these terms have no agreed-upon definition. People with their own favorite definition of "fixed and random effects" don't always realize that other definitions are out there. Worse, people conflate different definitions.

our experiment)

participants, stimuli, etc)

- parameters are the *same* across repeated measures
- sometimes call *population average effect*

- parameters *vary* for each "cluster" of repeated measures
- provides explicit model of dependence in the data

Linear *Mixed* Model: What's getting Mixed?

The inference we want to make about the population from our sample

Fixed Effects

- parameter estimates we typically care about just like normal regression (e.g. IVs in our experiment)
- parameters are the *same* across repeated measures
- sometimes call *population average effect*

The variation/uncertainty we want to generalize over when making that inference

Varying (Random) Effects

- the *variation* we want to take into account (e.g. differences between departments, participants, stimuli, etc)
- parameters *vary* for each "cluster" of repeated measures
- provides explicit model of dependence in the data

GLMs only estimates these!

The inference we want to make about the population from our sample

Fixed Effects

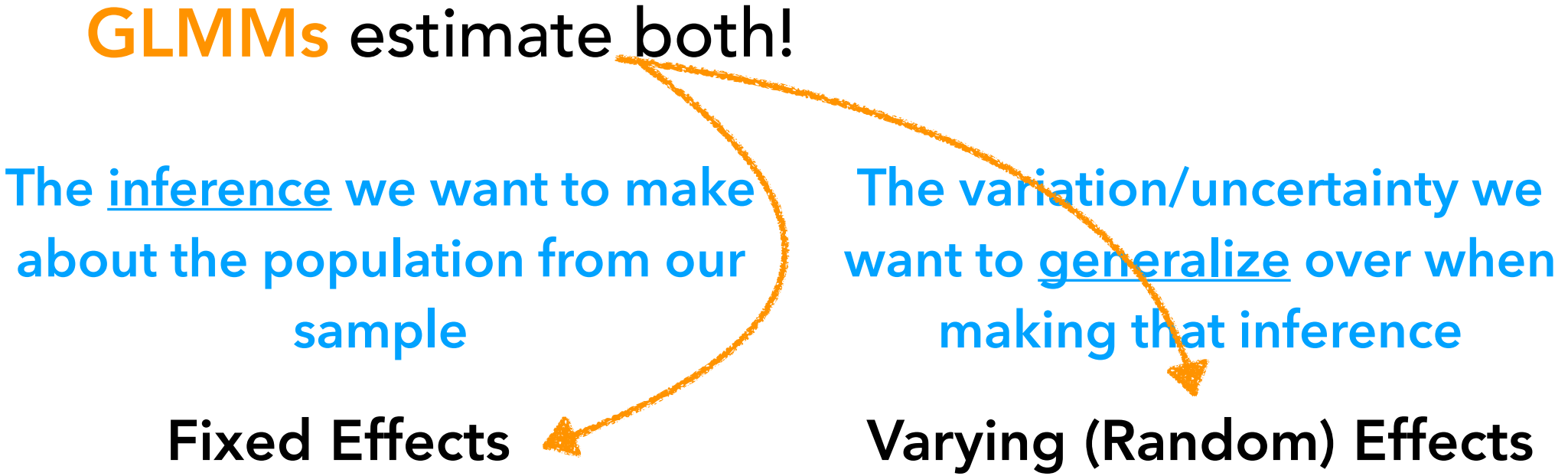
- parameter estimates we typically care about just like normal regression (e.g. IVs in our experiment)
- parameters are the *same* across repeated measures
- sometimes call *population average effect*

The variation/uncertainty we want to generalize over when making that inference

Varying (Random) Effects

- the *variation* we want to take into account (e.g. differences between departments, participants, stimuli, etc)
- parameters *vary* for each “cluster” of repeated measures
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GLMMs estimate both!



The inference we want to make about the population from our sample

Fixed Effects

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The variation/uncertainty we want to generalize over when making that inference

Varying (Random) Effects

- the *variation* we want to take into account (e.g. differences between departments, participants, stimuli, etc)
- parameters *vary* for each “cluster” of repeated measures
- provides explicit model of dependence in the data

Linear *Mixed* Model: What's getting Mixed?

The inference we want to make about the population from our sample

The variation/uncertainty we want to generalize over when making that inference

Fixed Effects

Varying (Random) Effects

"Reaction ~ Days + (1 + Days | Subject)"

What is the relationship between **Days** and **Reaction** in population?

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
309	0	222.73
309	1	205.27
309	2	202.98
309	3	204.71

Accounting for the fact that each **Subject** can have a different baseline **Reaction** and different effect of **Days**...

Linear *Mixed* Model: What's getting Mixed?

The inference we want to make about the population from our sample

The variation/uncertainty we want to generalize over when making that inference

Fixed Effects

Varying (Random) Effects

"Reaction ~ Days + (1 + Days | Subject)"

What is the relationship between Days and Reaction in population?

Accounting for the fact that each **Subject** can have a different baseline **Reaction** and different effect of **Days**...

Estimates inform each other and result in **shrinkage**

Linear *Mixed* Model: What's getting Mixed?

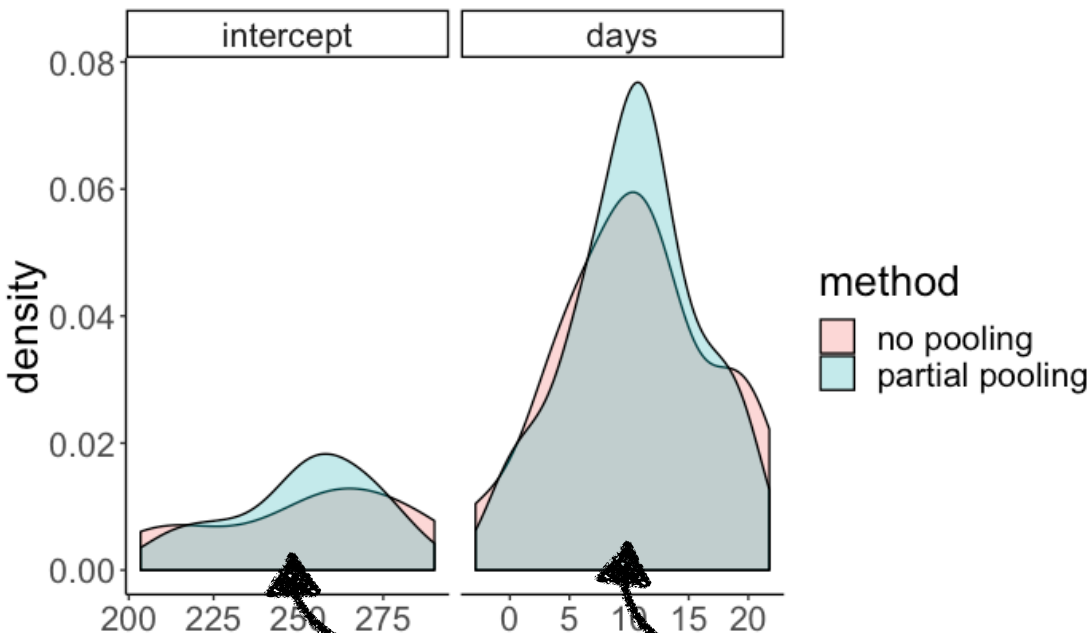
Fixed Effects/Factors

Varying Effects/Factors

"Reaction ~ Days + (1 + Days | Subject)"



Estimates inform each other
and result in **shrinkage**

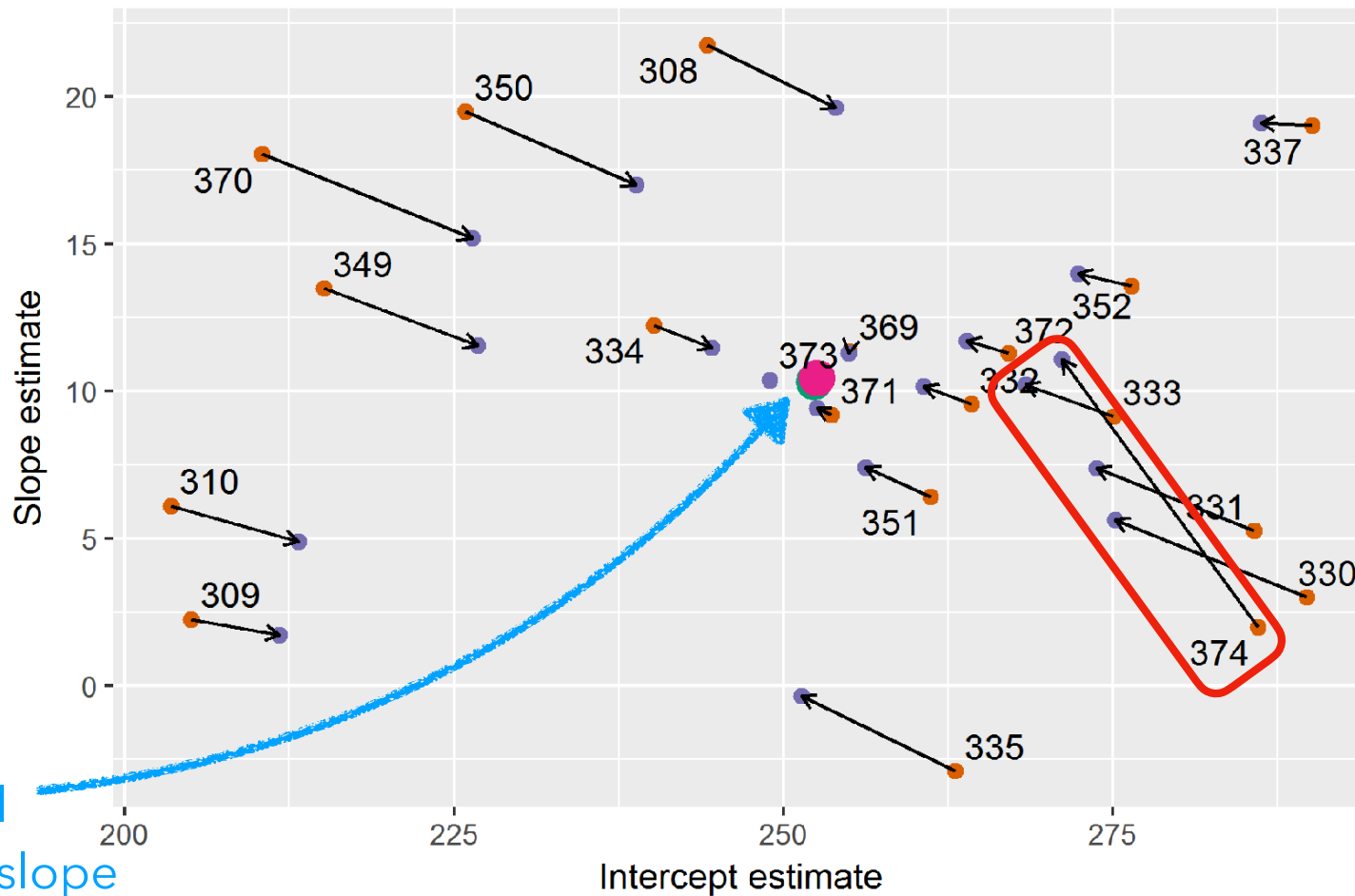


Variance "shrinks"

method	intercept	days
no pooling	28.95	6.56
partial pooling	21.59	5.46

Distributions over Subject

Estimates inform each other and result in shrinkage



we only had
two
observations

- More shrinkage when estimate is further from the average
- More shrinkage when estimate is more uncertain (based on fewer obs) so more information "borrowed" from other clusters

Estimates inform each other and result in **shrinkage**



The term “shrinkage” may have negative connotations. John Tukey preferred to refer to the process as the estimates for individual subjects “**borrowing strength**” from each other.

This is a **fundamental difference between mixed-effects models versus strictly fixed effects models**. In a mixed-effects model we assume that the levels of a grouping factor are a selection from a population and, as a result, can be expected to share characteristics to some degree. Consequently, the **predictions from a mixed-effects model are attenuated relative to those from strictly fixed-effects models**.

Douglas Bates (author of LMM library in R lme4)

Linear Mixed Model: What's happening under-the-hood?

GLM

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$Reaction_i = \beta_0 + \beta_1 Days_i + \epsilon_i$$

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
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309	1	205.27
309	2	202.98

i = single observation/row

Linear Mixed Model: What's happening under-the-hood?

GLM

$$y_i = \underline{\beta_0} + \underline{\beta_1}X_i + \epsilon_i$$

$$Reaction_i = \beta_0 + \beta_1 Days_i + \epsilon_i$$

$i =$ single observation/row

Fixed effects

(G)LMM

$$y_{ij} = (\underline{\beta_0} + b_{0j}) + (\underline{\beta_1} + b_{1j})X_{ij} + \epsilon_{ij}$$

Linear Mixed Model: What's happening under-the-hood?

GLM

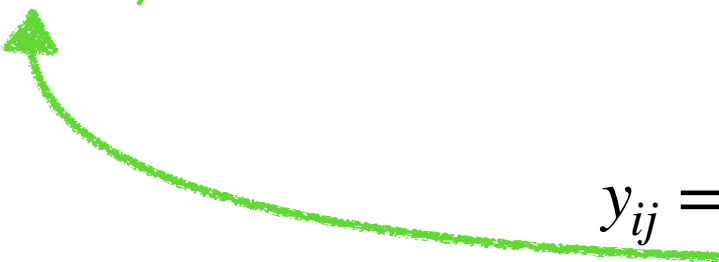
$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$Reaction_i = \beta_0 + \beta_1 Days_i + \epsilon_i$$

$i =$ single observation/row

Varying effects
(offsets)

(G)LMM

$$y_{ij} = (\beta_0 + \underline{b_{0j}}) + (\beta_1 + \underline{b_{1j}})X_{ij} + \epsilon_{ij}$$


Linear Mixed Model: What's happening under-the-hood?

GLM

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$\text{Reaction}_i = \beta_0 + \beta_1 \text{Days}_i + \epsilon_i$$

$i =$ single observation/row

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$$y_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})X_{ij} + \epsilon_{ij}$$

$$\text{Reaction}_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})\text{Days}_{ij} + \epsilon_{ij}$$

$ij =$ i observation for subject j

Linear Mixed Model: What's happening under-the-hood?

GLM

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$$\text{Reaction}_i = \beta_0 + \beta_1 \text{Days}_i + \epsilon_i$$

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(G)LMM

$$y_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})X_{ij} + \epsilon_{ij}$$

$$\text{Reaction}_{ij} = (\beta_0 + \underline{b_{0j}}) + (\beta_1 + \underline{b_{1j}})\text{Days}_{ij} + \epsilon_{ij}$$

Subject j 's
intercept & slope

$ij =$ i observation for subject j

Linear Mixed Model: What's happening under-the-hood?

(G)LMM

$$y_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})X_{ij} + \epsilon_{ij}$$

$$Reaction_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})Days_{ij} + \epsilon_{ij}$$

Really 2 equations (levels)

General relationship (fixed)

$$Y_{sd} = \beta_{0s} + \beta_{1s}X_{sd} + e_{sd} \quad \text{Level 1}$$

Participant level (varying)

$$\begin{aligned} \beta_{0s} &= \gamma_0 + S_{0s} \\ \beta_{1s} &= \gamma_1 + S_{1s} \end{aligned} \quad \text{Level 2}$$

Variance across Intercepts
Variance across Slopes

$$\langle S_{0s}, S_{1s} \rangle \sim N(\langle 0, 0 \rangle, \Sigma)$$

$$\Sigma = \begin{pmatrix} \tau_{00}^2 & \rho\tau_{00}\tau_{11} \\ \rho\tau_{00}\tau_{11} & \tau_{11}^2 \end{pmatrix}$$

Error for each obs for each S

$$e_{sd} \sim N(0, \sigma^2)$$

Linear Mixed Model: What's happening under-the-hood?

(G)LMM

$$Reaction_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})Days_{ij} + \epsilon_{ij}$$

Variable	Type	Description
Y_{sd}	observed	Value of Reaction for subject s on day d
X_{sd}	observed	Value of days_deprived (0-7) for subject s on day d
β_{0s}	derived	level 1 intercept parameter
β_{1s}	derived	level 1 slope parameter
e_{sd}	derived	Error for subject s , day d
γ_0	fixed	Grand intercept ("gamma")
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S_{0s}	derived	Random intercept (offset) for subject s
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Σ	random	Variance-covariance matrix
τ_{00}^2	random	Variance of random intercepts
ρ	random	Random correlation between intercepts and slopes
τ_{11}^2	random	Variance of random slopes
σ^2	random	Error variance

Level 1

$$Y_{sd} = \beta_{0s} + \beta_{1s}X_{sd} + e_{sd}$$

Level 2

$$\beta_{0s} = \gamma_0 + S_{0s}$$

$$\beta_{1s} = \gamma_1 + S_{1s}$$

Variance Components

$$\langle S_{0s}, S_{1s} \rangle \sim N(\langle 0, 0 \rangle, \Sigma)$$

$$\Sigma = \begin{pmatrix} \tau_{00}^2 & \rho\tau_{00}\tau_{11} \\ \rho\tau_{00}\tau_{11} & \tau_{11}^2 \end{pmatrix}$$

$$e_{sd} \sim N(0, \sigma^2)$$

Linear Mixed Model: What's happening under-the-hood?

- We estimate **fixed effects** from **data**

Variable	Type	Description
Y_{sd}	observed	Value of Reaction for subject s on day d
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β_{0s}	derived	level 1 intercept parameter
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$$e_{sd} \sim N(0, \sigma^2)$$

Linear Mixed Model: What's happening under-the-hood?

- We estimate fixed effects from data
- We estimate **variance & covariance** of **random-effects**

Variable	Type	Description
Y_{sd}	observed	Value of Reaction for subject s on day d
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$$e_{sd} \sim N(0, \sigma^2)$$

Linear Mixed Model: What's happening under-the-hood?

- We estimate fixed effects from data
- We estimate variance & covariance of random-effects
- Then we can derive (not estimate) **Subject offsets from fixed effects**

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Y_{sd}	observed	Value of Reaction for subject s on day d
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Linear Mixed Model: What's happening under-the-hood?

- We estimate fixed effects from data
- We estimate variance & covariance of random-effects
- Then we can derive (not estimate) Subject offsets from fixed effects

Varying (random) effects are the variability around **Fixed effects**

This is *how* LMMs **explicitly model dependence** in data

Variable	Type	Description
Y_{sd}	observed	Value of Reaction for subject s on day d
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β_{0s}	derived	level 1 intercept parameter
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LMMs: The Essentials

1. Ways to aggregate dependent data
 - Complete, No-pooling, **Partial-pooling**
2. Parts of the mixed model
 - Fixed vs Varying Effects
 - What's happening under-the-hood
3. Specifying Varying Effects
 - Random Intercepts & Slopes
 - How to choose your varying effects

Specifying Varying Effects

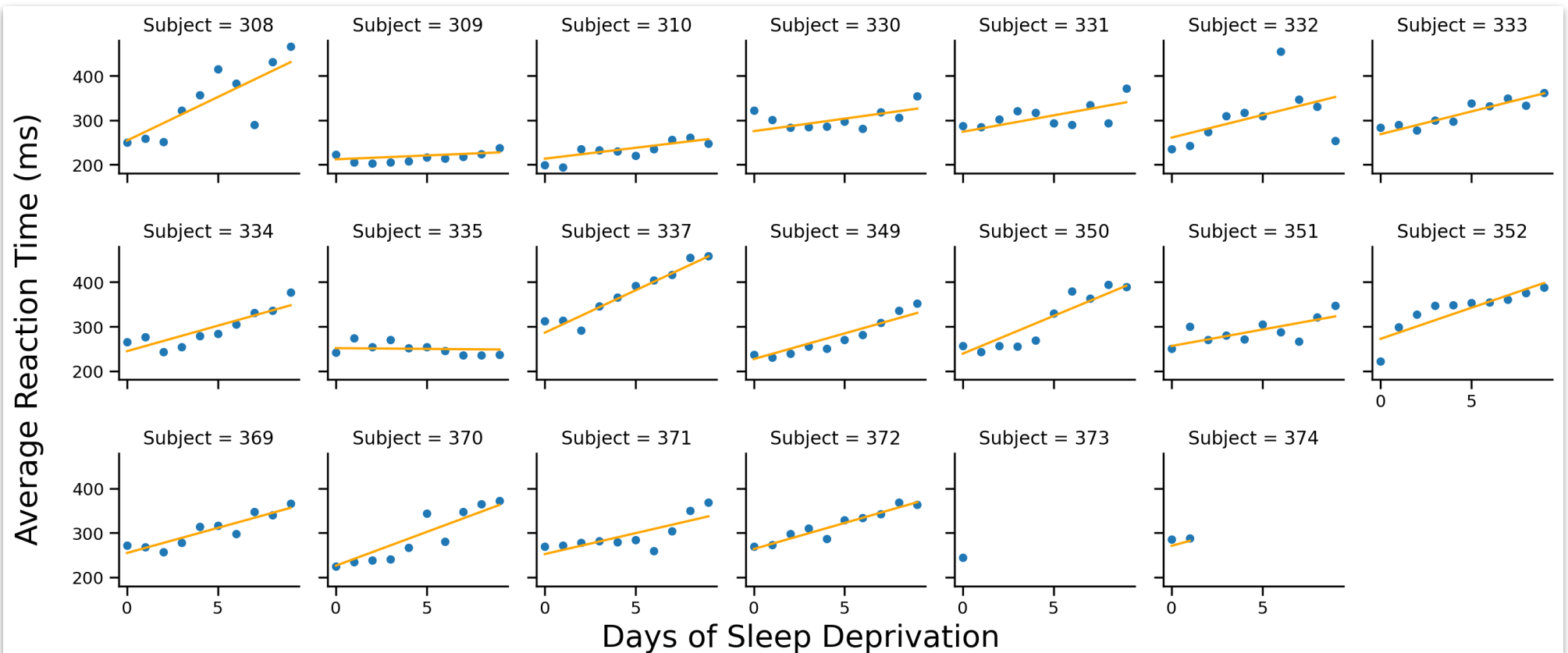
intercept

slope

```
model("Reaction ~ Days + (1 + Days | Subject)",  
data = sleep).fit()
```

how to model dependence

Each Subject is assumed to **independently deviate**
from a global **Intercept & Slope**

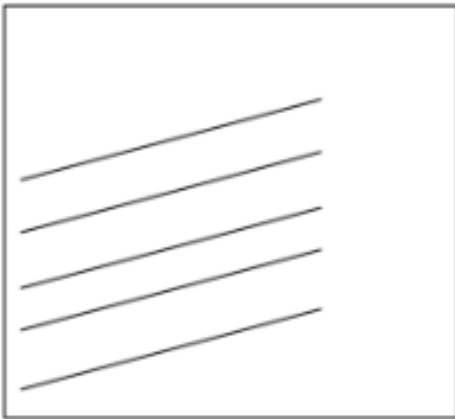


Specifying Varying Effects

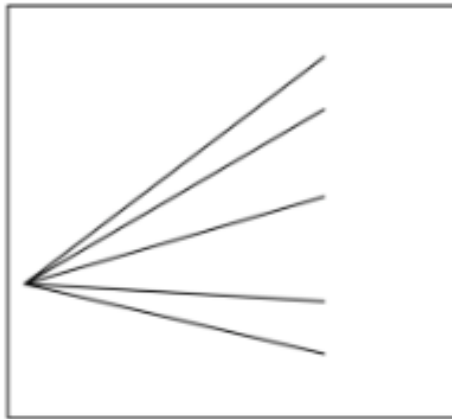
But we can model dependence in **multiple ways**

Each Subject is assumed to have unique **Intercept** & **Slope**

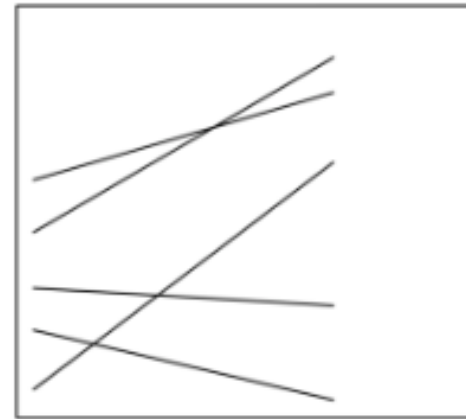
Varying intercepts



Varying slopes

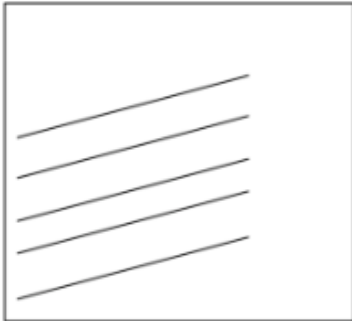


Varying intercepts and slopes

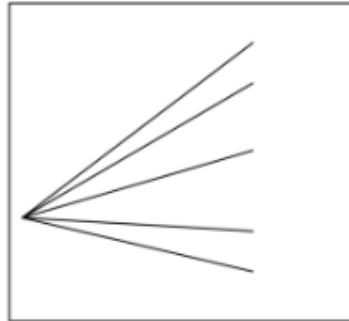


Specifying Varying Effects

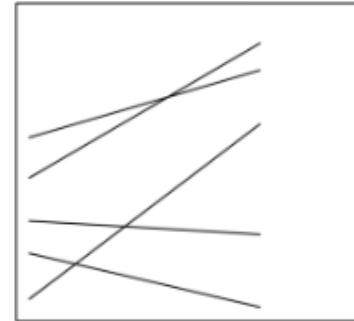
Varying intercepts



Varying slopes



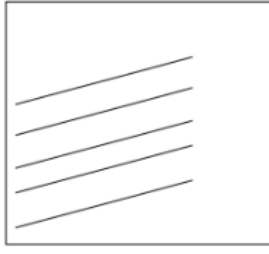
Varying intercepts and slopes



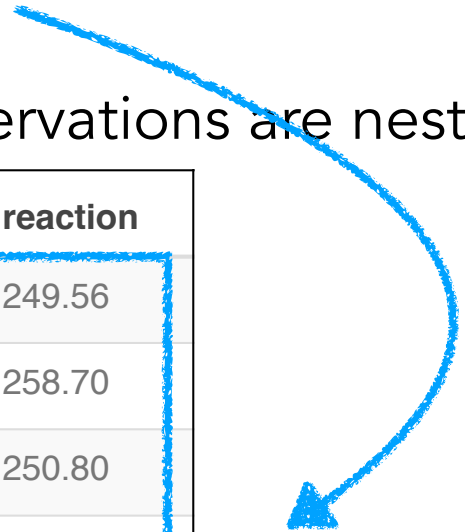
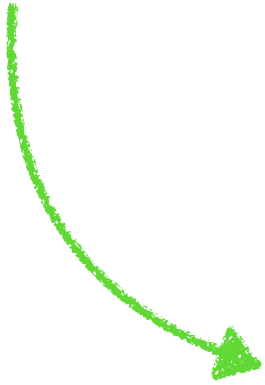
subject	days	reaction
308	0	249.56
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309	3	204.71
309	4	207.72

Specifying Varying Effects

Varying intercepts



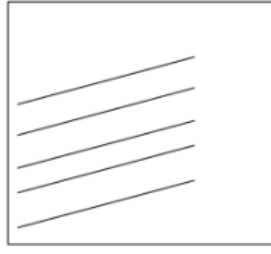
- Random **Intercept**
 - Captures **baseline variation in DV** for each cluster of replicates
 - Replicates = the observations
 - Cluster = what repeated observations are nested inside of



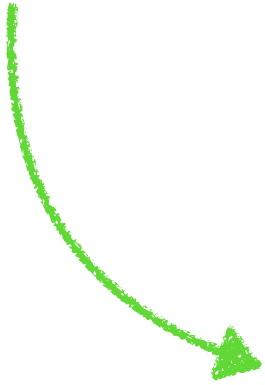
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Specifying Varying Effects

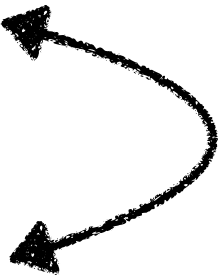
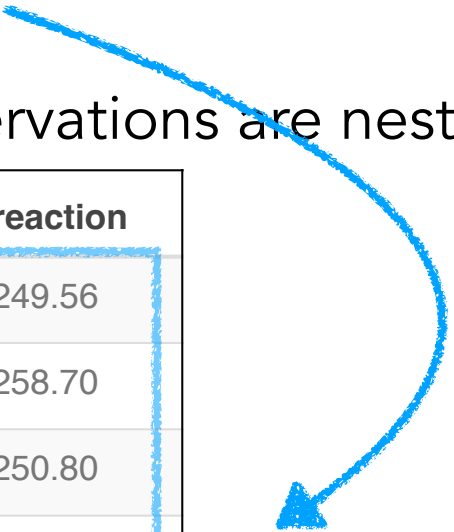
Varying intercepts



- Random **Intercept**
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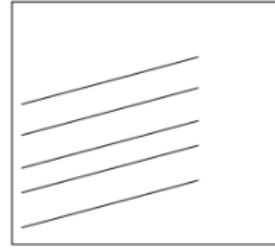
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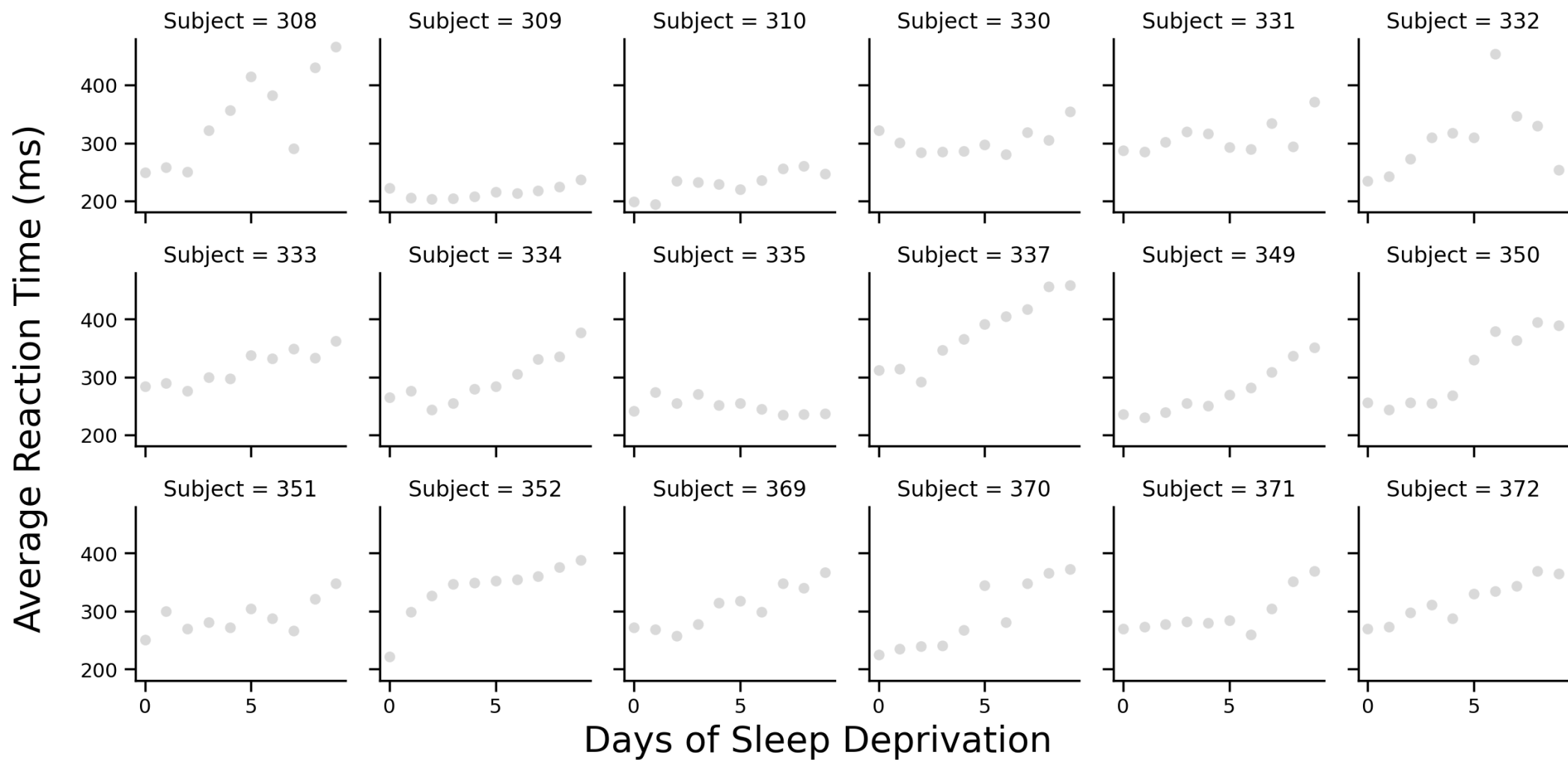
Account for the fact that these subjects have different average RTs

Specifying Varying Effects

Varying intercepts

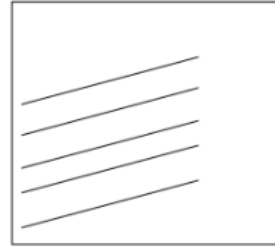


- Random **Intercept**
 - Captures **baseline variation in DV** for each cluster of replicates

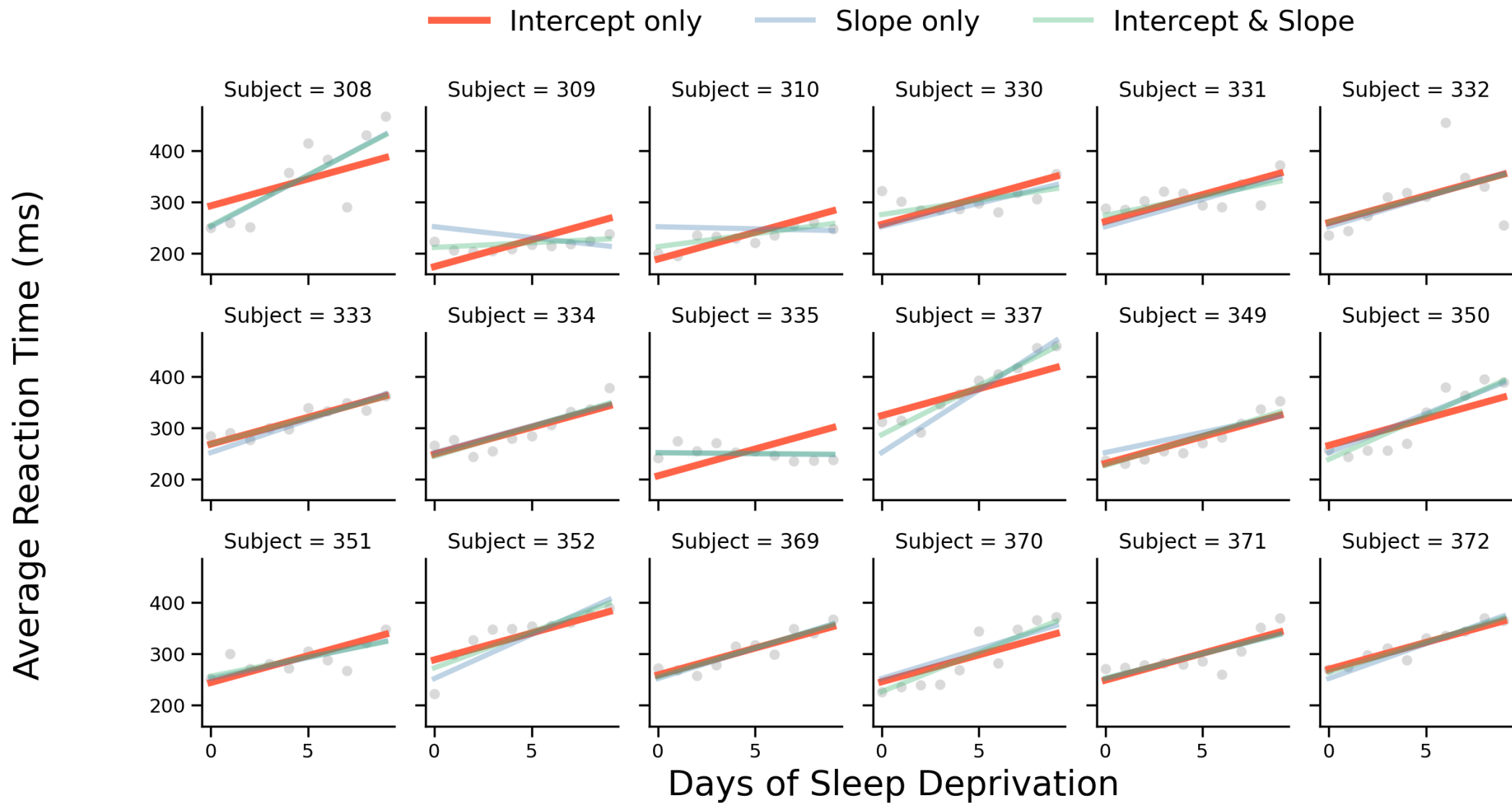


Specifying Varying Effects

Varying intercepts

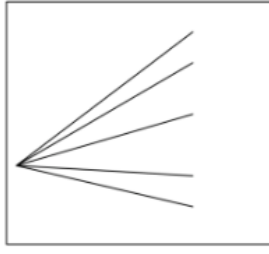


- Random **Intercept**
- Captures **baseline variation in DV** for each cluster of replicates



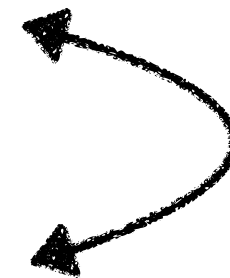
Specifying Varying Effects

Varying slopes



- Random **Slope**
- Captures **variation in effect of IV(s)** for each cluster of replicates

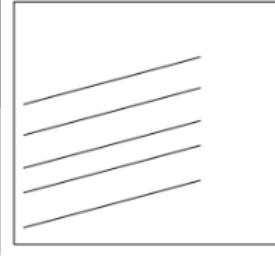
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308	1	258.70
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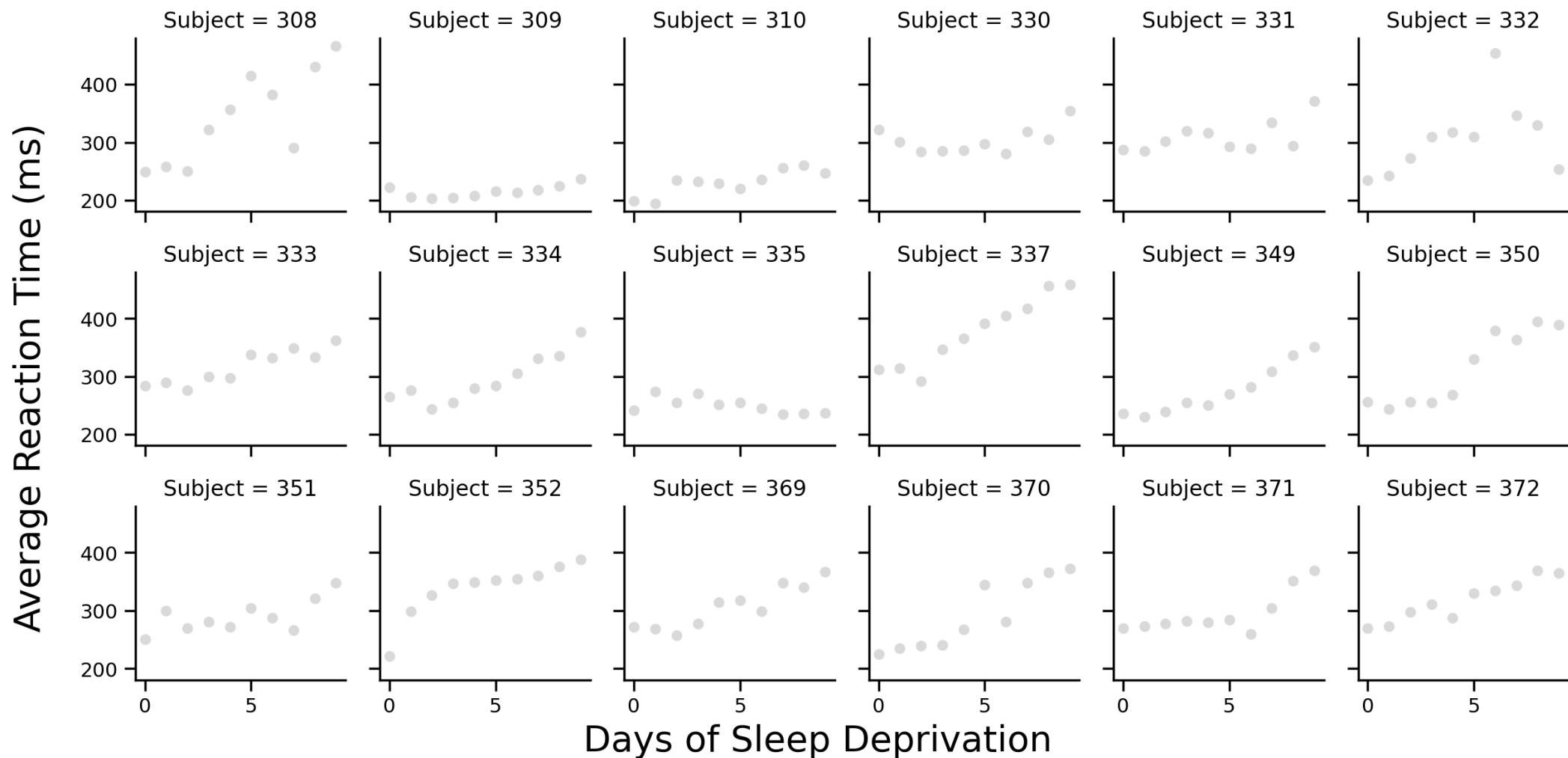
Account for the fact that these subjects have different relationships between RT ~ day

Specifying Varying Effects

Varying intercepts

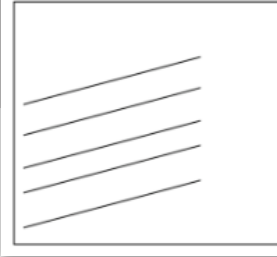


- Random **Slope**
 - Captures **variation in effect of IV(s)** for each cluster of replicates

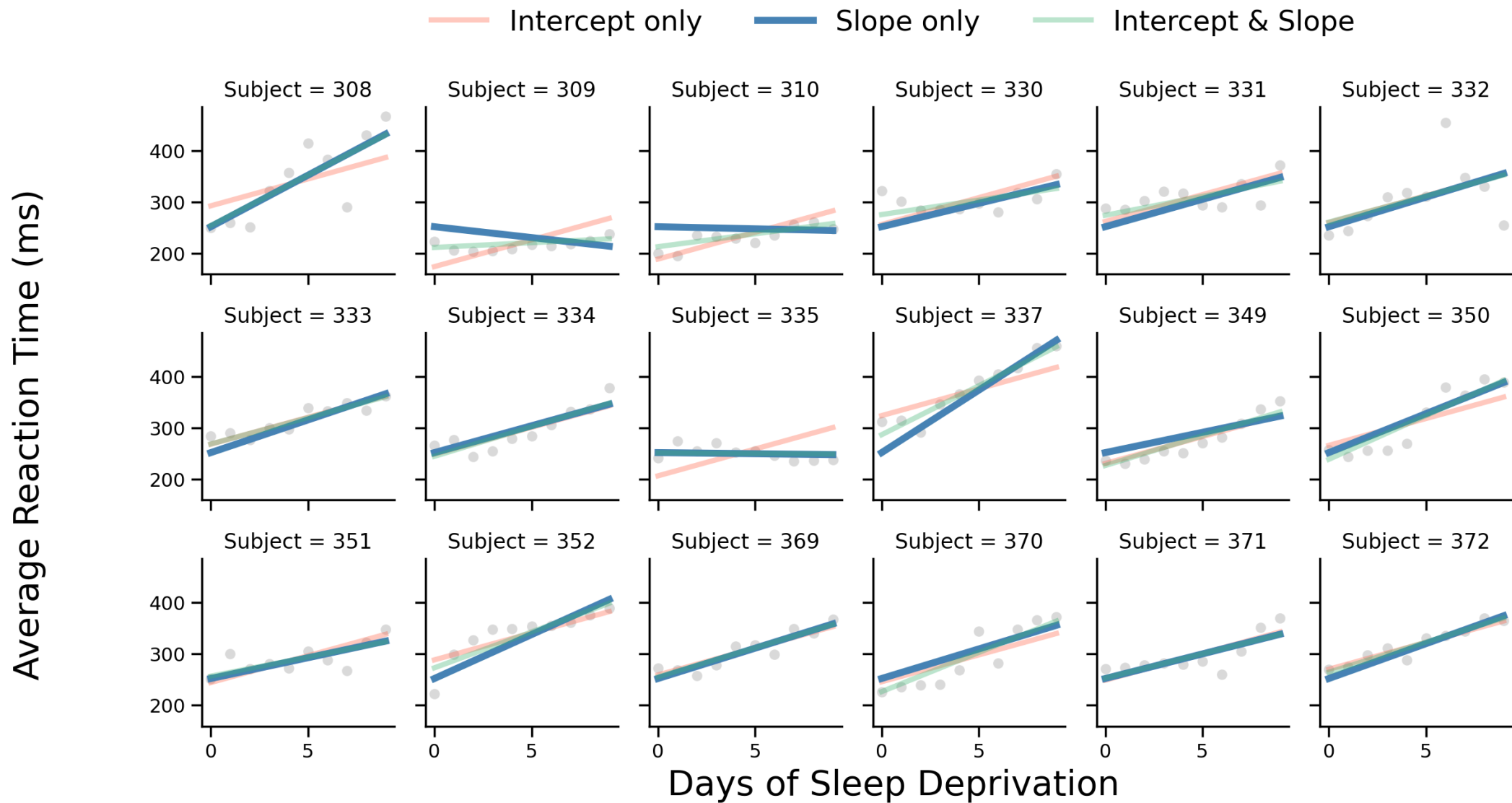


Specifying Varying Effects

Varying intercepts

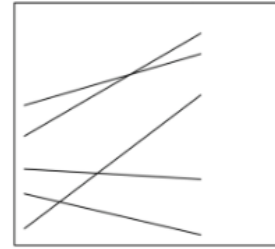


- Random **Slope**
- Captures **variation in effect of IV(s)** for each cluster of replicates



Specifying Varying Effects

Varying intercepts and slopes

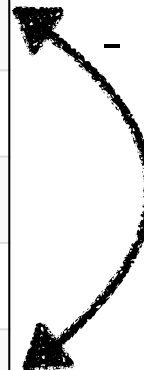


- Random **Intercept + Slope** (& typically correlation)
- Captures **variation in both baseline DV, effect of IV(s), and their correlation** for each cluster of replicates

subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
309	0	222.73
309	1	205.27
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309	3	204.71
309	4	207.72

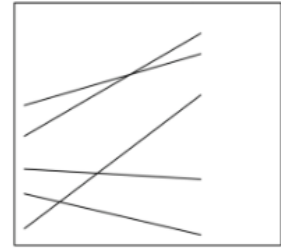
Account for:

- different baseline RTs
- different relationships between RT ~ day
- different correlation between baseline RT and RT ~ day
 - e.g. larger RTs = steeper slope
 - *slower* Subjects show *stronger* effect of Sleep Deprivation

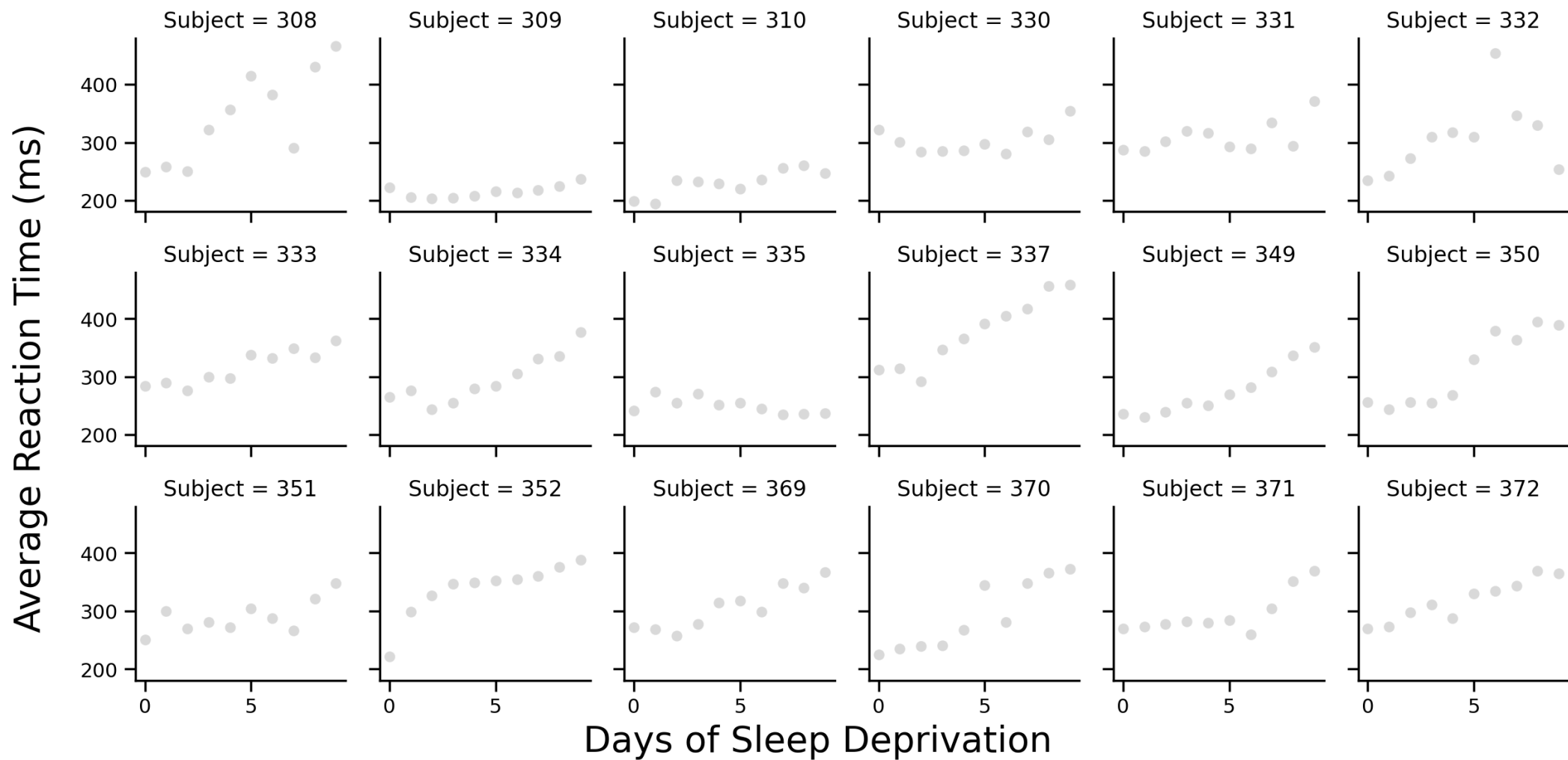


Specifying Varying Effects

Varying intercepts and slopes

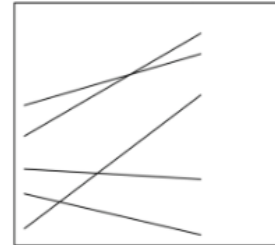


- Random **Intercept + Slope** (& typically correlation)
- Captures **variation in both baseline DV, effect of IV(s), and their correlation** for each cluster of replicates



Specifying Varying Effects

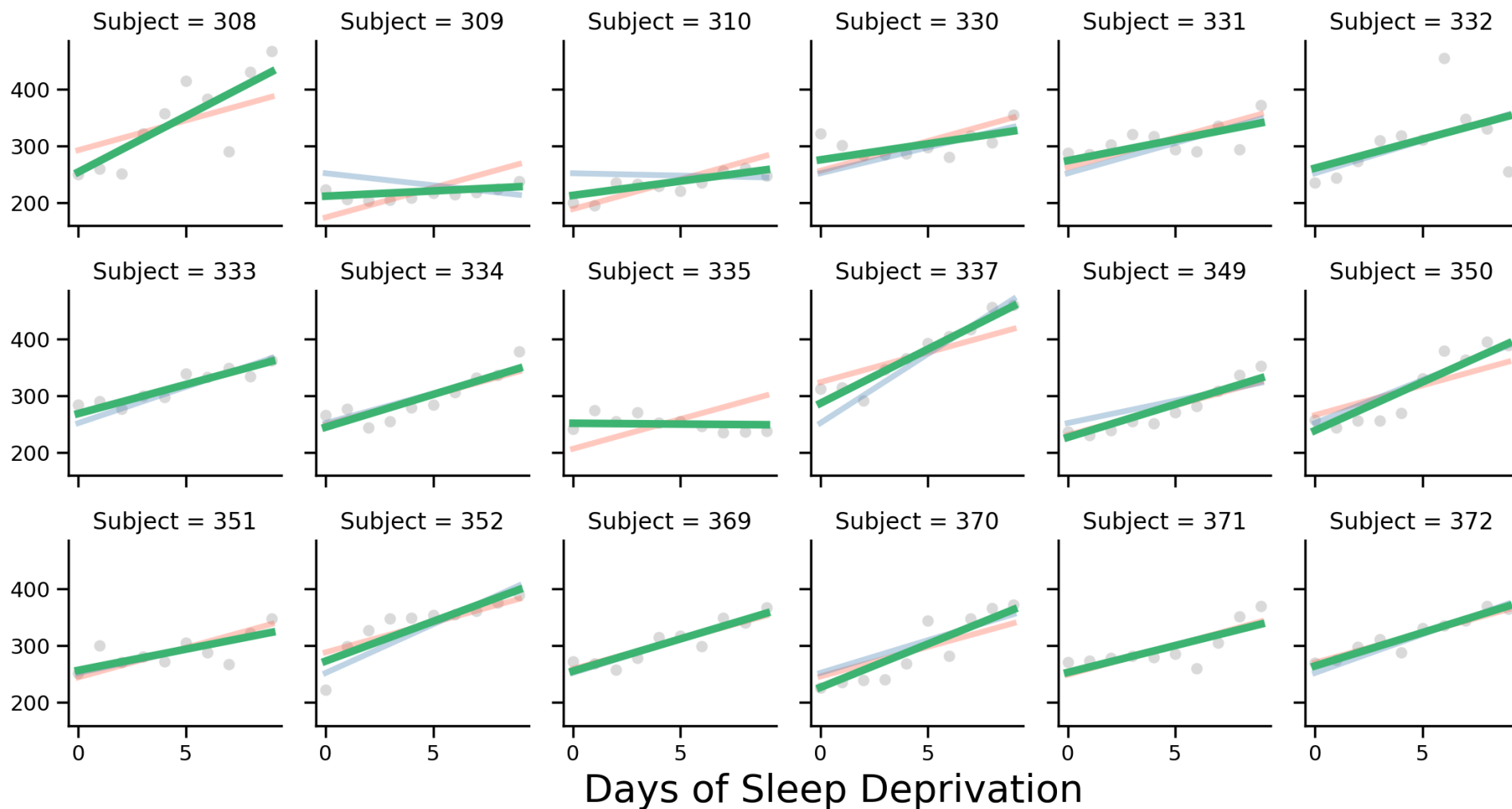
Varying intercepts and slopes



- Random **Intercept + Slope** (& typically correlation)
- Captures **variation in both baseline DV, effect of IV(s), and their correlation** for each cluster of replicates

— Intercept only — Slope only — Intercept & Slope

Average Reaction Time (ms)



What should I include as Varying Effects?

General Advice: **Start Maximal** then prune down



Contents lists available at [SciVerse ScienceDirect](#)

Journal of Memory and Language

journal homepage: www.elsevier.com/locate/jml

Random effects structure for confirmatory hypothesis testing:
Keep it maximal

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^b Department of Linguistics, University of California at San Diego, La Jolla, CA 92093-0108, USA

^c Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

*"Through theoretical arguments and Monte Carlo simulation, we show that LMMs generalize best when they include the **maximal random effects structure justified by the design...***

***Maximal LMMs should be the 'gold standard' for confirmatory hypothesis testing in psycholinguistics and beyond.**"*

- Failure to include maximal random-effect structures in LMMs (when such random effects are present in the underlying populations) **inflates Type I error rates**
- For designs including within-subjects (or within-items) manipulations, random-**intercepts-only LMMs can have catastrophically high Type I error rates**, regardless of how p-values are computed from them
- In terms of power, maximal models perform surprisingly well even in a "worst case" scenario where they assume random slope variation that is actually not present in the population

What should I include as Varying Effects?

General Advice: **Start Maximal** then prune down

- Start with random intercepts, slopes, & correlations for each repeated factor you want to generalize over (e.g. Subject, Stimulus Category, etc)
- Use **model comparison** to see whether extra varying effects parameters are **worth it**
- **Reduce complexity** of the varying effects structure **step-by-step**
 - Remove the correlation between varying effects first, then slopes, last intercepts
- **Examine the variance** across your clusters
 - Sometimes you don't have enough variance to estimate random effect; model will **fail to converge**

Tomorrow's' lab notebook(s) will cover these!

Another Important Consideration: Centering

- With GLMMs we have 2 options for how to center
 - **Group-mean centering** (Level 1): with cluster (Subject)
 - **Grand-mean centering** (Level 2): across all data

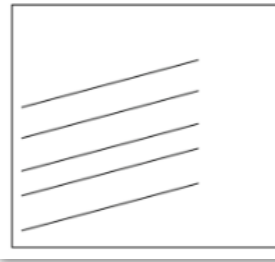
You have think a bit more carefully

We've added a bunch of resources to course-website

Raw uncentered variables	Grand-mean centering	Group-mean centering
• Rarely makes sense unless there is a desire to estimate intercept and intercept variance when the predictor is equal to zero (e.g., mean or variance of group-means for females only). • Does not make sense when $X = 0$ is an unreasonable or undesirable value (e.g., age = 0) • Entails assumption that group-means are uncorrelated with the predictor (Algina & Swaminathan, 2011).	• For level-2 variables controlling for level-1 variables. • If adjusted group-mean is desired interpretation of average intercept, and variance of adjusted group-means is desired interpretation of intercept variance. • Interactions between level-2 variables. (Interaction estimate and test are unaffected, but lower order terms are). • Entails assumption that group-means are uncorrelated with predictor. • Inclusion of level-2 variables in a model without level-1 reintroduced mean variables will not fully control for the group-level effects of individual-level variables.	• Use if pure level-1 effect is desired without considering level-2 variables. Group-mean centered variables will be uncorrelated with level-2 variables and therefore estimates of the effect of the level-1 variable will not partial out level-2 variables. • When accurate slope variance estimates are desired • When relative between and within effects of the same construct at level 1 and level 2 is desired (e.g., SES and average class SES). • If effects of level-2 variables only of interest without regard to partialling level-1 variability out. • When cross-level interactions are of interest and interpretation of "main effect" is of interest. • Inclusion of level-2 variables in a model without level-1 reintroduced mean variables will not fully control for the covariate.

Estimating LMMs: Syntax & Coefficients

Varying intercepts



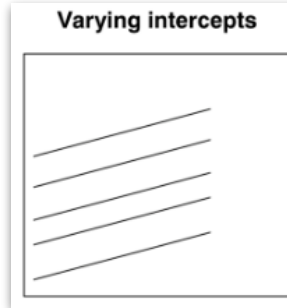
Random Intercept

```
model("Reaction ~ Days + (1 | Subject)",  
data = sleep)
```

Formula (Varying effects only)	Description
(1 Subject)	Random intercept for each Subject
(0 + X1 Subject)	Random slope for each Subject
(X1 Subject) (1 + X1 Subject)	Correlated random intercept & slope for each Subject
(X1 Subject) (1 Subject) + (0 + X1 Subject)	Uncorrelated random intercept & slope for each Subject
(X1 + X2 Subject)	Correlated random intercept & slope for each Subject for multiple predictors
(X1 * X2 Subject)	Same but include interaction too
(X1 Subject) + (X2 Stimulus)	Correlated intercepts and slopes for multiple repeated variables

Estimating LMMs: Syntax & Coefficients

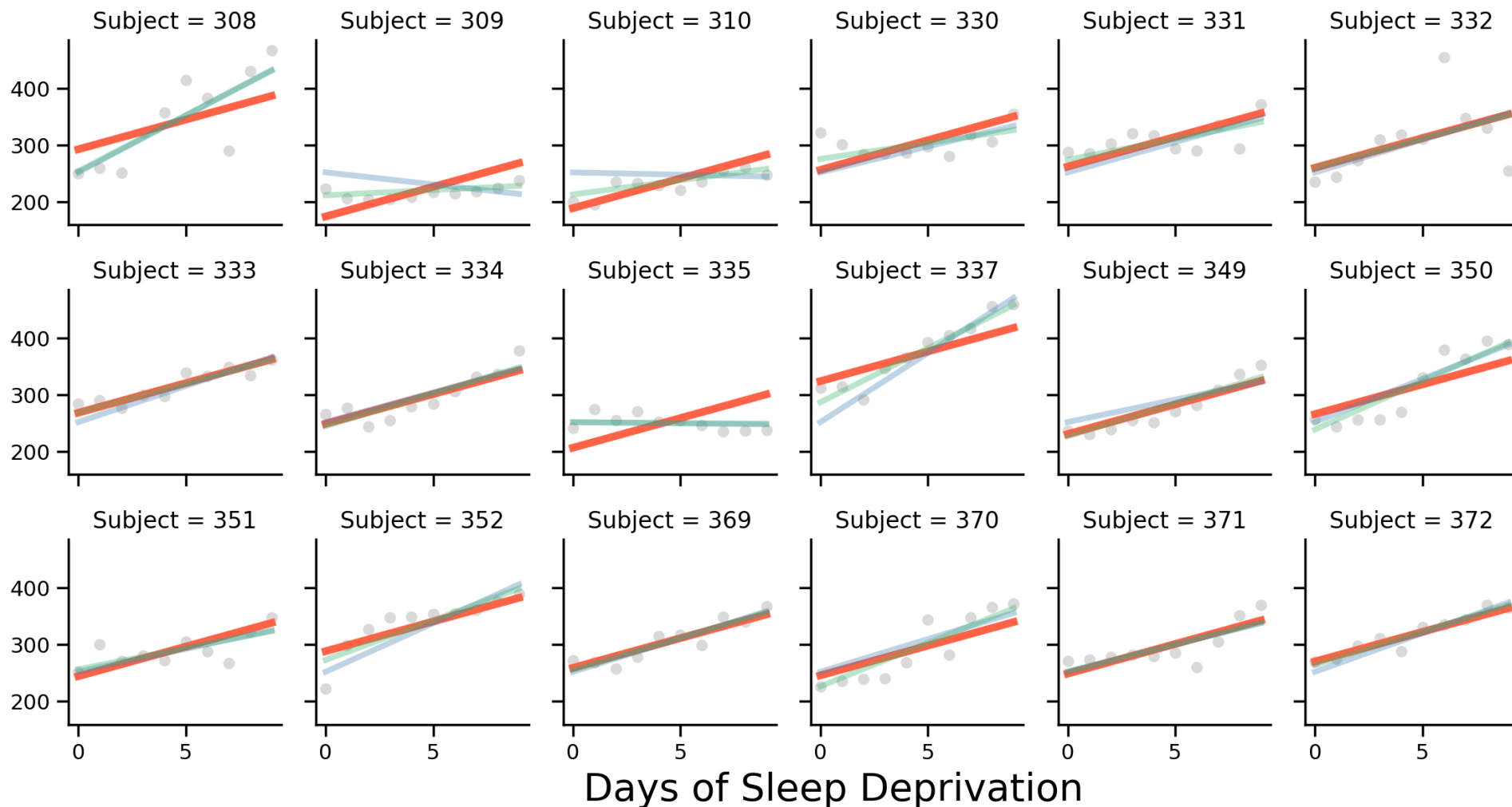
Random Intercept



```
model("Reaction ~ Days + (1 | Subject)",  
data = sleep)
```

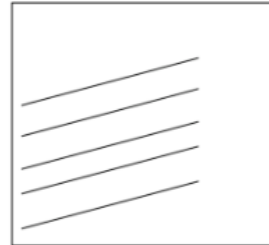
— Intercept only — Slope only — Intercept & Slope

Average Reaction Time (ms)



Estimating LMMs: Syntax & Coefficients

Varying intercepts



Random Intercept

```
model("Reaction ~ Days + (1 | Subject)",  
data = sleep)
```

Fixed Effects Parameters

	Estimate	2.5_ci	97.5_ci	SE	DF	T-stat	P-val	Sig
(Intercept)	251.405105	232.301892	270.508318	9.746716	22.810199	25.793826	2.241351e-18	***
Days	10.467286	8.891041	12.043531	0.804221	161.000000	13.015428	6.412601e-27	***

Remember the data

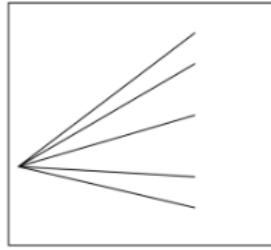
subject	days	reaction
308	0	249.56
308	1	258.70
308	2	250.80
308	3	321.44
308	4	356.85
309	0	222.73
309	1	205.27
309	2	202.98
309	3	204.71
309	4	207.72

	(Intercept)	Days
308	292.188815	10.467286
309	173.555551	10.467286
310	188.296537	10.467286
330	255.811547	10.467286
331	261.621294	10.467286
332	259.626343	10.467286
333	267.905599	10.467286
334	248.408124	10.467286
335	206.122978	10.467286
337	323.587791	10.467286
349	230.208856	10.467286
350	265.516468	10.467286
351	243.542884	10.467286

Varying Effects Parameters
Different Intercept per Subject

Estimating LMMs: Syntax & Coefficients

Varying slopes



Random Slope

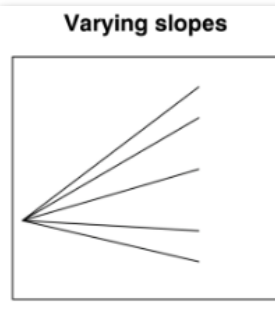
```
model("Reaction ~ Days + (0 + Days | Subject)",  
data = sleep)
```

Formula (Varying effects only)	Description
(1 Subject)	Random intercept for each Subject
(0 + X1 Subject)	Random slope for each Subject
(X1 Subject) (1 + X1 Subject)	Correlated random intercept & slope for each Subject
(X1 Subject) (1 Subject) + (0 + X1 Subject)	Uncorrelated random intercept & slope for each Subject
(X1 + X2 Subject)	Correlated random intercept & slope for each Subject for multiple predictors
(X1 * X2 Subject)	Same but include interaction too
(X1 Subject) + (X2 Stimulus)	Correlated intercepts and slopes for multiple repeated variables

Estimating LMMs: Syntax & Coefficients

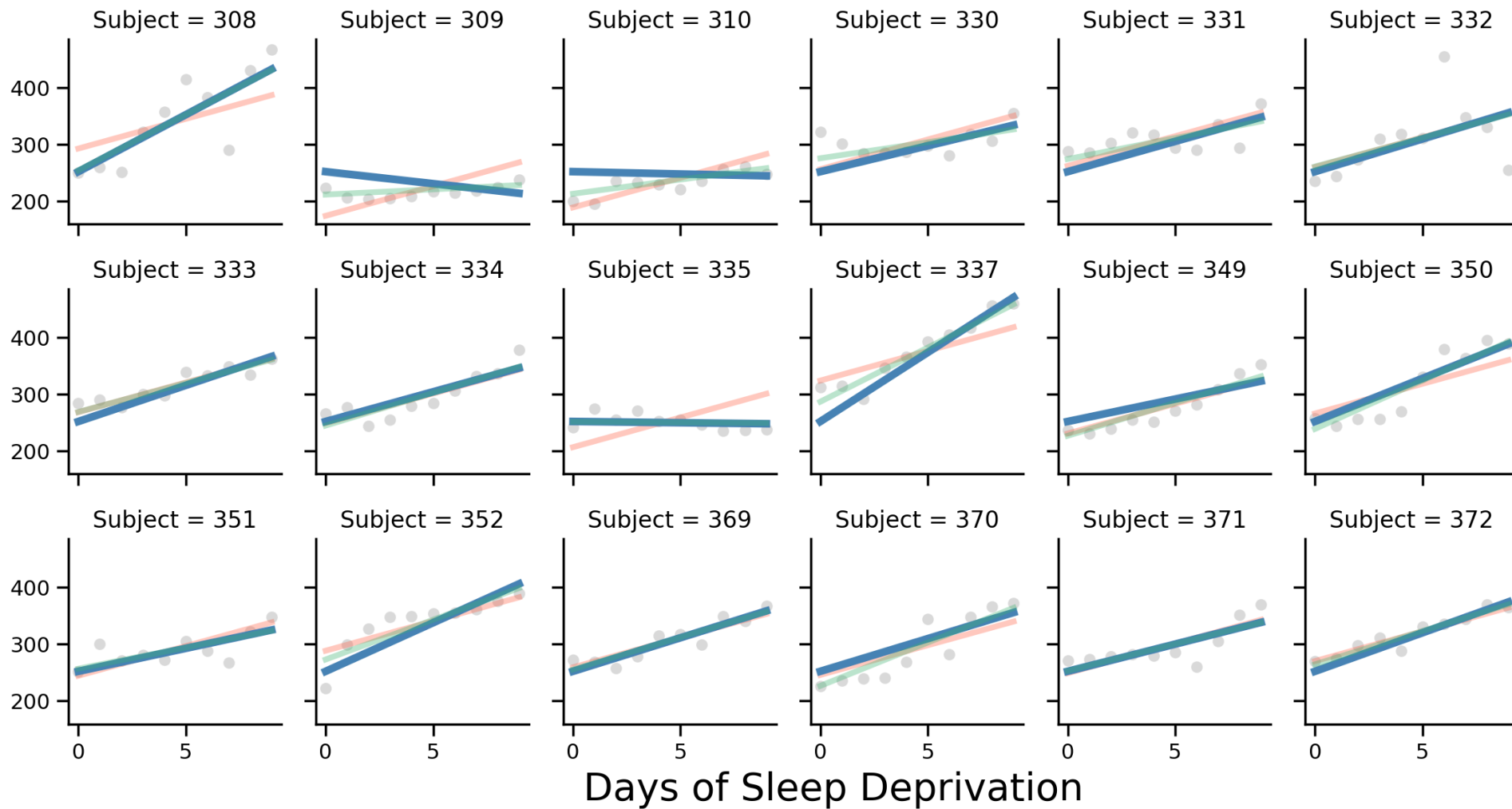
Random Slope

```
model("Reaction ~ Days + (0 + Days | Subject)",  
data = sleep)
```



— Intercept only — Slope only — Intercept & Slope

Average Reaction Time (ms)



Estimating LMMs: Syntax & Coefficients

Random Slope

```
model("Reaction ~ Days + (0 + Days | Subject)",  
data = sleep)
```

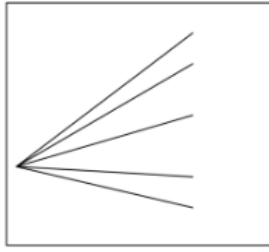
Fixed Effects Parameters

	Estimate	2.5_ci	97.5_ci	SE	DF	T-stat	P-val	Sig
(Intercept)	251.405105	243.526107	259.284103	4.019971	161.000003	62.539039	7.333827e-115	***
Days	10.467286	6.803019	14.131552	1.869558	21.679342	5.598802	1.319902e-05	***

	(Intercept)	Days
308	251.405105	20.086692
309	251.405105	-4.232671
310	251.405105	-0.818920
330	251.405105	9.127388
331	251.405105	10.675484
332	251.405105	11.535298
333	251.405105	12.743009
334	251.405105	10.477487
335	251.405105	-0.433739
337	251.405105	24.357732
349	251.405105	7.906925
350	251.405105	15.201214
351	251.405105	8.104156

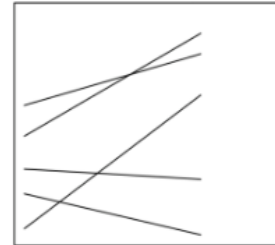
Varying Effects Parameters
Different Slopes per Subject

Varying slopes



Estimating LMMs: Syntax & Coefficients

Varying intercepts and slopes



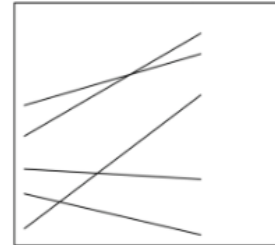
Random Intercept, Slope, & Correlation

```
model("Reaction ~ Days + (1 + Days | Subject)",  
data = sleep)
```

Formula (Varying effects only)	Description
(1 Subject)	Random intercept for each Subject
(0 + X1 Subject)	Random slope for each Subject
(X1 Subject) (1 + X1 Subject)	Correlated random intercept & slope for each Subject
(X1 Subject) (1 Subject) + (0 + X1 Subject)	Uncorrelated random intercept & slope for each Subject
(X1 + X2 Subject)	Correlated random intercept & slope for each Subject for multiple predictors
(X1 * X2 Subject)	Same but include interaction too
(X1 Subject) + (X2 Stimulus)	Correlated intercepts and slopes for multiple repeated variables

Estimating LMMs: Syntax & Coefficients

Varying intercepts and slopes



Random **Intercept**, **Slope**, & **Correlation**

OR:

```
model("Reaction ~ Days + (Days | Subject)",  
data = sleep)
```

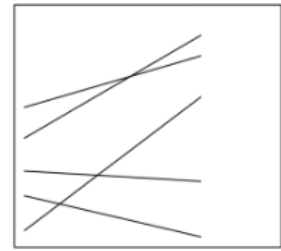
Formula (Varying effects only)	Description
(1 Subject)	Random intercept for each Subject
(0 + X1 Subject)	Random slope for each Subject
(X1 Subject) (1 + X1 Subject)	Correlated random intercept & slope for each Subject
(X1 Subject) (1 Subject) + (0 + X1 Subject)	Uncorrelated random intercept & slope for each Subject
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(X1 Subject) + (X2 Stimulus)	Correlated intercepts and slopes for multiple repeated variables

Estimating LMMs: Syntax & Coefficients

Random **Intercept**, **Slope**, & **Correlation**

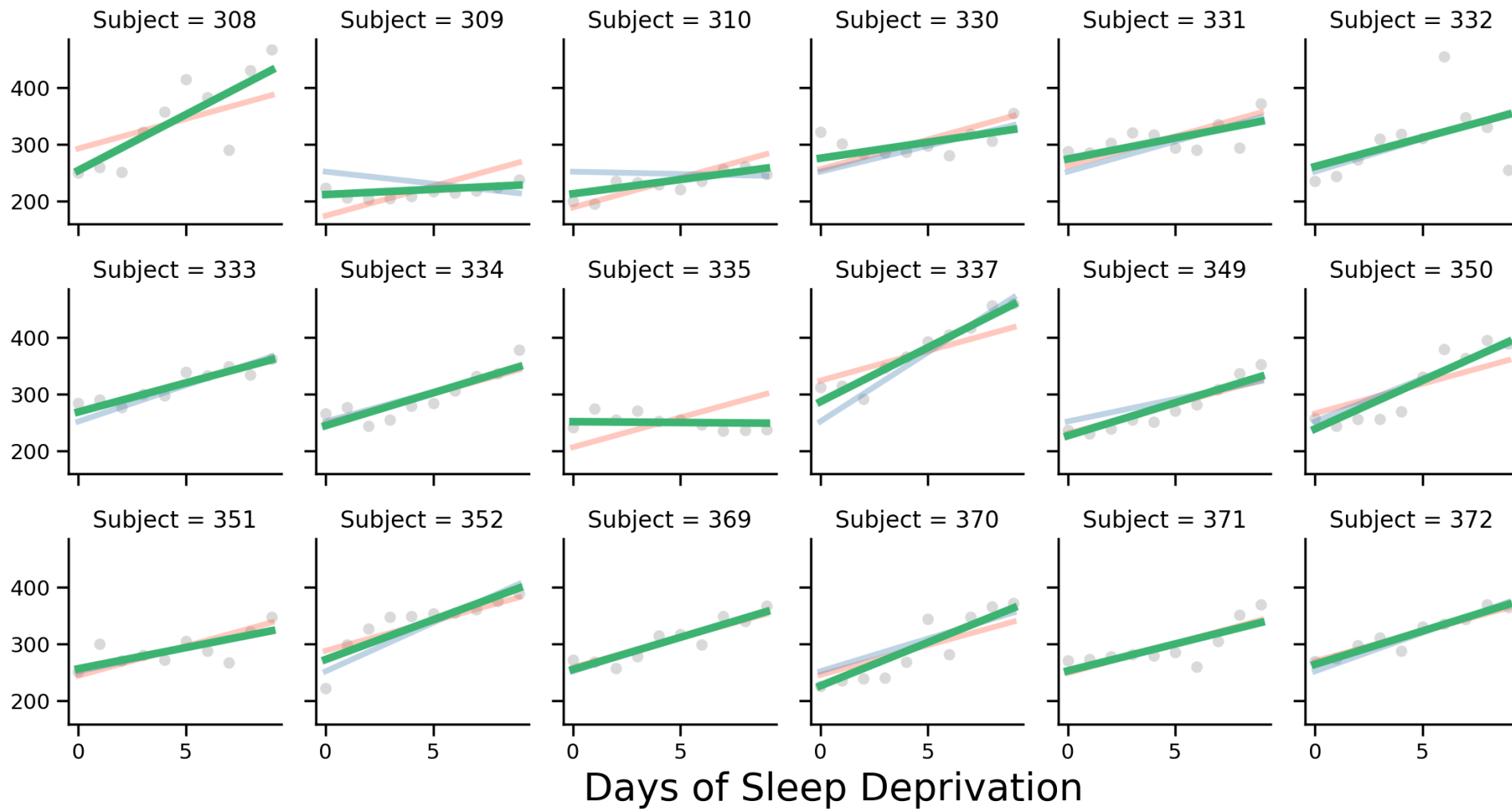
```
model("Reaction ~ Days + (Days | Subject)",  
data = sleep)
```

Varying intercepts and slopes



— Intercept only — Slope only — Intercept & Slope

Average Reaction Time (ms)



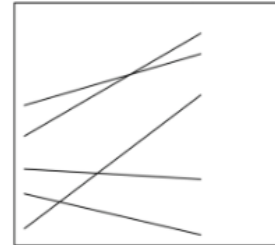
Estimating LMMs: Syntax & Coefficients

Random Intercept, Slope, & Correlation

```
model("Reaction ~ Days + (Days | Subject)",  
data = sleep)
```

Fixed Effects Parameters

Varying intercepts and slopes



	Estimate	2.5_ci	97.5_ci	SE	DF	T-stat	P-val	Sig
(Intercept)	251.405105	238.029141	264.781069	6.824597	16.999726	36.838090	1.171558e-17	***
Days	10.467286	7.437594	13.496978	1.545790	16.999984	6.771481	3.263824e-06	***

	(Intercept)	Days
308	253.663656	19.666262
309	211.006367	1.847605
310	212.444696	5.018429
330	275.095724	5.652936
331	273.665417	7.397374
332	260.444673	10.195109
333	268.245613	10.243650
334	244.172490	11.541868
335	251.071436	-0.284879
337	286.295592	19.095551
349	226.194876	11.640718
350	238.335071	17.081504
351	255.982969	7.452024

Varying Effects Parameters
Different Intercept **AND** Slopes
by Subject

Estimating LMMs: Syntax & Coefficients

Syntax conventions for specifying varying effects

Formula (Varying effects only)	Description
<code>(1 Subject)</code>	Random intercept for each Subject
<code>(0 + X1 Subject)</code>	Random slope for each Subject
<code>(X1 Subject)</code> <code>(1 + X1 Subject)</code>	Correlated random intercept & slope for each Subject
<code>(X1 Subject)</code> <code>(1 Subject) + (0 + X1 Subject)</code>	Uncorrelated random intercept & slope for each Subject
<code>(X1 + X2 Subject)</code>	Correlated random intercept & slope for each Subject for multiple predictors
<code>(X1 * X2 Subject)</code>	Same but include interaction too
<code>(X1 Subject) + (X2 Stimulus)</code>	Correlated intercepts and slopes for multiple repeated variables

Estimating LMMs: Syntax & Coefficients

You can also specify random effects for multiple variables

We'll explore more in notebooks

Formula (Varying effects only)	Description
<code>(1 Subject)</code>	Random intercept for each Subject
<code>(0 + X1 Subject)</code>	Random slope for each Subject
<code>(X1 Subject)</code> <code>(1 + X1 Subject)</code>	Correlated random intercept & slope for each Subject
<code>(X1 Subject)</code> <code>(1 Subject) + (0 + X1 Subject)</code>	Uncorrelated random intercept & slope for each Subject
<code>(X1 + X2 Subject)</code>	Correlated random intercept & slope for each Subject for multiple predictors
<code>(X1 * X2 Subject)</code>	Same but include interaction too
<code>(X1 Subject) + (X2 Stimulus)</code>	Correlated intercepts and slopes for multiple repeated variables

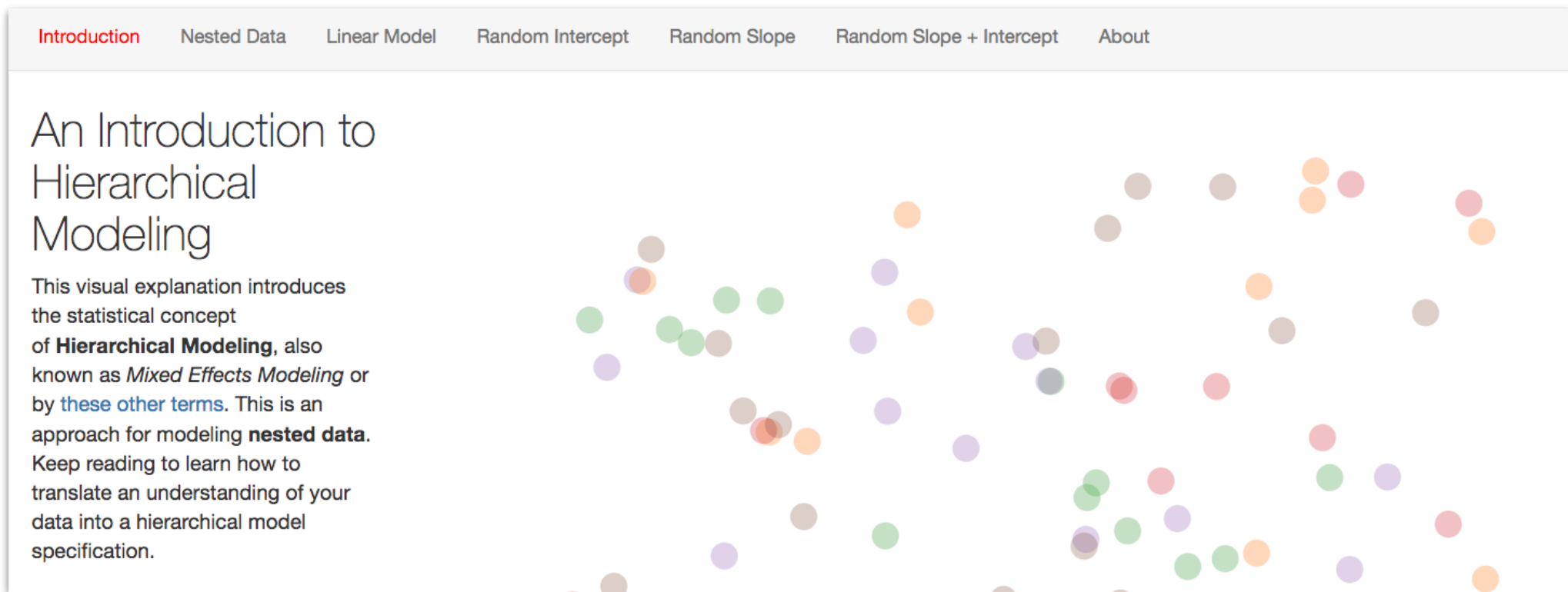
LMMs: Summary

An extension of GLM that allows us to **account for dependencies in data** by **estimating parameters at different levels**

1. Ways to aggregate dependent data
 - Complete, No-pooling, **Partial-pooling**
2. Parts of the mixed model
 - **Fixed** vs **Varying** Effects
 - What's happening under-the-hood
3. Specifying Varying Effects
 - Random **Intercepts** & **Slopes**
 - How to choose your varying effects

With remaining time check out:

<https://mfviz.com/hierarchical-models/>



Additional References on course website!